Security Risks and the Screening of Investments*

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Abstract

Governments increasingly restrict foreign investment, arguing that foreign participation in sensitive industries can threaten national security. We develop a dynamic model of how a host government should manage exposure to foreign firms under geopolitical uncertainty. The government benefits economically from foreign investment but faces the risk that some firms are covertly adversarial and may build capacity—through data access, supply-chain dependence, or similar channels—to inflict harm, such as espionage or economic disruption, if relations with the firm's home country become hostile. We characterize the government's optimal access policy, accounting for the adversarial firm's strategic incentives to build capacity. When geopolitical relations are initially peaceful but may later turn hostile, optimal exposure is U-shaped over time: the government initially grants high access, reduces it as the period of potential hostility approaches, and raises it again if hostility does not occur. This pattern contrasts with the seemingly intuitive policy of restricting access initially and relaxing restrictions after a probationary period. We show that such a probationary policy is instead optimal when the government expects geopolitical relations to potentially turn hostile in the immediate future.

Keywords: Dynamic Screening, Information Asymmetry, Geopolitical Risk, Foreign Investment, National Security, Economic Statecraft.

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1 Introduction

Foreign firms can become deeply integrated into the host country's economy, supplying capital, technology, and essential services. However, this integration also creates vulnerabilities: access to sensitive data or control over critical supplies may be used against the host country if relations with the firm's home country turn hostile. Governments use investment screening mechanisms – regulatory procedures that allow governments to review, condition, or block foreign investments on national security grounds – to manage these concerns. Examples include U.S. restrictions on Huawei,[1] and the 2016 decision to block a proposed Chinese acquisition of the U.S. business of Aixtron, a German semiconductor equipment manufacturer.[2] Such restrictions, however, sacrifice the economic gains from foreign investment. This raises the central question we examine in this paper: how should a government manage the access of foreign firms to its economy over time under geopolitical uncertainty?

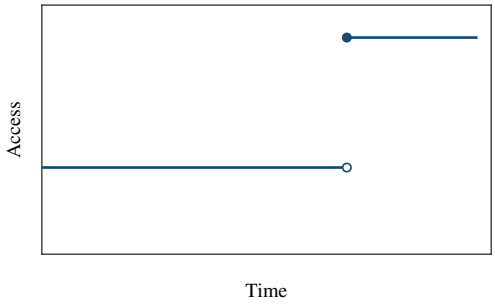
The answer depends on understanding how access affects the firm's ability to act against the host country and the government's information about its intentions. A firm can use its access to build vulnerabilities that could be exploited if geopolitical relations turn hostile. It could accumulate sensitive information (Bauerle Danzman, 2021; Farrell and Newman, 2019), or deepen domestic firms' dependence on its products or services that are difficult to replace (Clayton, Maggiori, and Schreger, 2026; Pearson, Rithmire, and Tsai, 2022). However, the activities that create these vulnerabilities also give the government an opportunity to learn if a firm will undermine it during hostility (Gartzke and Lindsay, 2015; Maschmeyer, 2021). A firm seeking to exploit these vulnerabilities during hostility must, therefore, weigh the benefits of creating them against the risk of being identified.

We study this problem in a dynamic model in which a host government regulates its exposure to a foreign firm by choosing how much access to grant it over time. Access generates

[1] Jill C. Gallagher, *U.S. Restrictions on Huawei Technologies: National Security, Foreign Policy, and Economic Interests*, Congressional Research Service Report R47012, January 10, 2022, https://crsreports.congress.gov/product/pdf/R/R47012.

[2] Reuters, "China's Fujian Drops Aixtron Bid After Obama Blocks Deal," December 8, 2016, https://www.reuters.com/article/us-aixtron-m-a-fujian/chinas-fujian-drops-aixtron-bid-after-obama-blocks-deal-idUSKBN13X16H.

(a) Persistent geopolitical risk



(b) Anticipated geopolitical risk

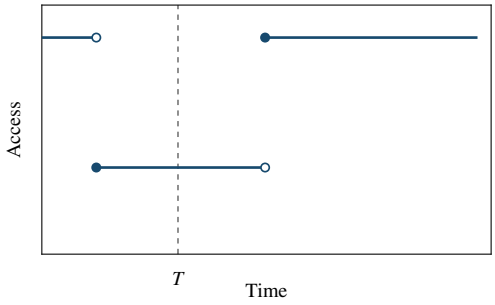


Figure 1: Optimal exposure paths. The dashed line in panel (b) marks the onset of geopolitical risk: hostility may arise only after T.

economic benefits, but a firm seeking to act against the host country can also use it to prepare for hostility. The firm chooses how much to prepare, weighing the benefits of greater capabilities against the risk of detection. We examine how optimal exposure depends on the timing of geopolitical risk. The onset of this risk marks when hostility becomes possible, while the arrival of hostility itself remains uncertain.

Our central result is that optimal exposure can be U-shaped when geopolitical risk begins only after an anticipated period of peace. As Figure 1(b) illustrates, the government initially grants full access, restricts it as the risky period approaches, and liberalizes again if peace continues. Initial openness is optimal even though it allows the firm to create greater vulnerabilities to exploit during future hostility. This contrasts with the starting-small logic in models of learning and screening (Hua and Watson, 2022; Kolb and Madsen, 2023; Rauch and Watson, 2003; Sobel, 1985), as well as related models of long-term relationships (Albuquerque and Hopenhayn, 2004; Atakan, Koçkesen, and Kubilay, 2020; Fudenberg, Georgiadis, and Rayo, 2021; Ghosh and Ray, 1996). Economic gains from greater initial access arrive earlier, while the security costs from that exposure arrive only in the future. The government therefore offers greater exposure initially, accepting greater vulnerability in the future in exchange for larger economic gains before the risk begins.

When geopolitical risk is persistent, hostility can arise at any moment from the beginning of the relationship. The optimal exposure follows the intuitive probationary pattern as illustrated in Figure 1(a): the government initially restricts access and later liberalizes if peace continues, even though the risk of hostility remains unchanged and the firm's intentions remain uncertain.

Eventual reopening reflects how the consequences of additional exposure change as the relationship evolves. Greater access generates economic benefits, but also allows the firm to enhance its abilities to inflict more harm during hostility. These preparations also create opportunities for the government to identify the firm as adversarial. The firm undertakes further preparations only when the benefits of being able to inflict more harm outweigh the costs of possible detection. Its choices therefore ensure that the information generated due to monitoring the accumulation of this additional exposure only partially offsets the security costs that exposure creates. However, as exposure increases, the security cost of additional access decreases enough for the economic gain to justify reopening. The government, therefore, liberalizes even though further exposure still worsens its expected payoff during hostility.

A key insight that follows from our analysis is that a longer anticipated period of peace can shorten the duration when access is restricted. Greater initial openness allows exposure to accumulate more quickly, bringing forward the point at which the economic gains from additional access justify reopening. Thus, as the duration of the anticipated peace increases, the government can begin restricting access later and liberalizes earlier. In particular, the U-shaped policy under anticipated risk entails a shorter restriction period than the probationary policy under persistent risk.

These results contribute most directly to the extensive literature on investment screening. Studies of investment screening in the United States emphasize domestic political pressures as a driving force behind the adoption and expansion of these regulations (Bauerle Danzman and Meunier, 2023; Canes-Wrone, Mattioli, and Meunier, 2020; Kang, 1997). The demand

for screening also depends on what countries stand to lose from foreign access, with technologically advanced economies showing stronger support for screening and a greater likelihood of adopting it (Chan and Meunier, 2022; Eichenauer, Dorsch, and Wang, 2021). We complement these explanations by examining how a government with given economic and security objectives should regulate access within an ongoing investment relationship.

The model builds on three determinants identified by this literature on investment screening as well as the broader political economy of foreign investment. The economic gains from exposure reflect the interests of domestic firms and other constituencies that benefit from foreign investment (Danzman, 2020; Kalinova, Palerm, and Thomsen, 2010; Pandya, 2014). Existing work explains governments' preference for economic openness through domestic interests and partisanship (Milner, 1997; Pinto, 2013), democratic institutions (Milner and Kubota, 2005; Pandya, 2014), and firms' demand for external financing (Danzman, 2020; Rajan and Zingales, 2003). Related studies show how political risk (Jensen, 2008; Li and Resnick, 2003; Wellhausen, 2015) and international commitments (Büthe and Milner, 2008) affect foreign investment.

The security costs resulting from the firm's preparation and hostile action capture concerns that foreign access can create vulnerabilities that may later be exploited against the host country (Bauerle Danzman, 2021; Farrell and Newman, 2019). More generally, economic relationships can create asymmetric dependence and political leverage alongside economic gains (Baldwin, 1985; Hirschman, 1980; Keohane and Nye Jr, 1973; Mastanduno, 1998). More recent work shows how states exploit their position within economic networks (Drezner, Farrell, and Newman, 2021; Farrell and Newman, 2019) and how security concerns increasingly shape trade and investment policy (Meunier and Nicolaïdis, 2019; Roberts, Choer Moraes, and Ferguson, 2019).

Finally, uncertainty about the firm's intentions reflects the difficulty of distinguishing commercially independent firms from those susceptible to home-state pressures. Evidence from China shows that even private firms can maintain significant political connections as it car-

ries substantial consequences for their financing (Li, Meng, et al., 2008; Pearson, Rithmire, and Tsai, 2022). State control can also make firms' commercial decisions more responsive to interstate political relations (Davis, Fuchs, and Johnson, 2019). A case in point is the role of private U.S. banks' coordinated efforts against Allende's incoming government in Chile (Aldunate, González, and Prem, 2024; Harmer, 2011).

Theoretically, this paper contributes to models of learning and screening in a long-term relationship. These models explain how limiting the relationship early on allows learning (Ghosh and Ray, 1996; Kranton, 1996; Rauch and Watson, 2003; Sobel, 1985) or sustain cooperation (Atakan, Koçkesen, and Kubilay, 2020; Fudenberg, Georgiadis, and Rayo, 2021; Fudenberg and Rayo, 2019; Garicano and Rayo, 2017). When players can secretly undermine over the course of their relationship, starting small can limit the harm while allowing screening (Kolb and Madsen, 2023; Zalke, 2026). Our results highlight an important qualification to the starting-small logic: optimal access depends on when the government anticipates the relationship to become hostile. When the government anticipates that hostility cannot arise until a later date, it may grant high access initially, even though this allows the firm to develop greater capabilities and inflict more harm when hostility arrives.

More broadly, the analysis in this paper relates to the literature on how strategic choices under imperfect information shape conflict. Private information affects bargaining and responses to threats (Bils and Spaniel, 2017; Fearon, 1995; Fey and Ramsay, 2007; Kenkel and Schram, 2024). Present commitments and investments can change future bargaining power and create incentives for preventive action (Bas and Coe, 2016; Benson and Smith, 2023; Debs and Monteiro, 2014; Powell, 2006), while actions that reveal information affect security (Bils and Smith, 2025; Coe and Vaynman, 2020; Smith, 2021). Our model applies these mechanisms to study the regulation of foreign investment: access generates economic gains while enabling the firm to develop greater capabilities for future hostility. The government's policy therefore shapes both the firm's incentives to develop these capabilities and what it can learn about the firm's intentions.

The rest of the paper proceeds as follows. Section 2 presents the model. Section 3 presents the analysis. Section 4 analyzes the case when the geopolitical situation can worsen suddenly leading to hostility. Section 5 analyzes a setup where the government anticipates an increase in geopolitical tension after a peaceful period. We conclude with a discussion of the implications of the results and future directions for research.

2 Model

There are two players: a host government (G) and a foreign firm (F). The foreign firm can be of two types $\theta \in \{g, b\}$, where $\theta = g$ denotes a commercial (good) firm and $\theta = b$ an adversarial (bad) firm that may act to undermine the host country.

Time is continuous, $t \geq 0$, and there can be two states of the world: peace (p) or hostility (h). The relationship starts in peace but can turn to hostility due to a geopolitical shock. In the baseline environment, which we call *persistent geopolitical risk*, the shock arrives randomly according to a Poisson process with hazard rate $\lambda > 0$. Once the state turns hostile, it is absorbing.

At each time t, the government decides how much access or exposure to provide to the firm: it can either terminate the relationship with the firm in which case the firm will be excluded from its economy forever from then on, or choose to continue the relationship, in which case, the government determines its exposure to the firm. The feasible exposure level depends on the geopolitical state. During peace, the government chooses $x_t^p \in [\phi, 1]$; during hostility, it chooses $x_t^h \in [\phi_h, 1]$, where $0 < \phi_h \leq \phi < 1$. The exposure x_t^j, where $j \in p, h$, can be understood as a "normalized exposure index" that captures the foreign firm's effective integration into the domestic economy in terms of the corresponding economic benefits to the host country. Accordingly, $x_t^j = \phi_j$ and $x_t^j = 1$ denote the feasible minimum and maximum exposure, respectively, compatible with the geopolitical state j. We assume that ϕ_h can be arbitrarily small but positive.[3] Termination at any point effectively sets the future exposure

[3] Allowing ϕ_h to be very small reflects most settings where regulators can feasibly constrain foreign firms'

to 0. Both players discount future payoffs at rate $r > 0$. We assume the government has commitment power within each geopolitical state, but not across states. This reflects a setting in which the screening regulations are binding on the government, but the publicly observable onset of hostility permits it to revise them.

During peace, an adversarial firm can use the exposure to build capacity. At time t in peace, it chooses a capacity-building effort $\alpha_t \in [0, 1]$, so its accumulated capacity, C_t, which we denote as capacity stock, evolves according to $\dot{C}_t = \alpha_t x_t^p$. The commercial firm does not accumulate any capacity. For ease of calculations, we set $\alpha = 0$ for a commercial firm. The capacity building does not hurt the government directly. Thus, its flow payoff during peace is

$$\pi_t^p(x_t^p) = x_t^p.$$

The firm receives the same peace-time flow payoff x_t^p, irrespective of its type.

Once hostility arrives at date T_h, the capacity stock freezes and remains fixed at C_{T_h}. An adversarial firm can now use this capacity to secretly undermine the government. It chooses an undermining intensity $\beta_t \in [0, 1]$ that inflicts a flow loss of $(1 + KC_{T_h})\beta_t x_t^h$ on the host government, where $K > 0$ and x_t^h is the exposure during hostility at t. The multiplier $k(C_{T_h}) := 1 + KC_{T_h}$ denotes the effective harm per unit of exposure and undermining. Therefore, the flow payoffs for the government during hostility conditional on the firm being adversarial with capacity stock C_{T_h} and undermining intensity β_t are

$$\pi_t^h(x_t^h, \beta_t; C_{T_h}) = x_t^h - (1 + KC_{T_h})\beta_t x_t^h.$$

Here, the baseline effective harm equals the marginal economic gain from exposure, which is 1. We relax this equality in Appendix C, allowing the baseline harm to differ from the marginal economic gain by considering a more general specification $k(C) = a + KC$.[4]

integration without fully terminating the relationship. In practice, this can be done by limiting the scope of the firms' activities, such as capping interfaces and data permissions, restricting any expansion of the firm.

[4]The exposure patterns remain qualitatively unchanged under relatively mild restrictions on baseline harm and the prior probability that the firm is commercial, as characterized in Proposition C.8.

We consider a zero-sum game between the government and the adversarial firm during hostility. Therefore, the adversarial firm's hostility flow payoffs are $-\pi_t^h(x_t^h, \beta_t; C_{T_h})$. The payoffs for the government and the commercial firm are aligned even during hostility, and therefore, the commercial firm does not undermine. We can, therefore, set $\beta = 0$ for the commercial firm to reflect this. If the firm is excluded at any point, both the government and the firm receive an outside option of 0. Exclusion therefore prevents any accumulated capacity from generating further losses for the government. This reflects our focus on security threats whose realization requires firm's continued access in the host country during hostility.

Information. The firm's type is private information. The government's belief at time t that the firm is commercial is $q_t \in [0, 1]$, with initial belief $q_0 \in (0, 1)$. During peace, whenever the adversarial firm builds capacity with effort α_t over an interval dt, the government receives a private and unverifiable signal about its type with probability $\mu\alpha_t x_t^p dt$, where $\mu > 0$ denotes the effectiveness of the government's peace-time monitoring. Note that detection is not cumulative in past capacity building i.e., only the current capacity building generates signals. During hostility, each act of undermining is discovered at rate $\gamma\beta_t dt$, where $\gamma > 0$ reflects the quality of monitoring. The government does not observe its payoff until the end of the game.

Contracts and Timing. The government's access schedule and exclusion policy within a geopolitical state $j \in \{p, h\}$ can be denoted by a contract $\mathcal{C}^j = (x^j, T_R^j)$, where $x^j = (x_t^j)_t$ specifies the exposure while access remains active and T_R^j denotes the time when the firm's access is terminated. The relationship begins in peace, when the government offers a single peace-time pooling contract, $\mathcal{C}^p$. Upon receiving this contract, an adversarial firm chooses its capacity-building effort. When the geopolitical shock arrives, capacity accumulation ends, the peace-time contract expires, and the government commits to a new contract, $\mathcal{C}^h$, specifying the firm's exposure and termination during hostility. The adversarial firm then chooses its undermining intensity until the relationship is terminated. Termination ends the game, and each player collects their payoff.

Discussion of Modeling Choices

Proportionality Between Economic Benefits and Loss. The model assumes that the loss from undermining is proportional to the economic benefits generated by the exposure. Such a specification follows a standard payoff structure in models of long-term relationships where both the benefits and losses are proportional to the stakes of the relationship (Hua and Watson, 2022; Kolb and Madsen, 2023). Substantively, the same channels through which exposure to the firm generates benefits for the host country also provide opportunities for an adversarial firm to undermine it. Thus, this modeling specification provides a natural interpretation in which the potential loss from exposure is measured in terms of its economic benefits.

Uncertainty about Firm. In the model, the host government remains uncertain about the firm's type even when the home country's intentions are clear. Foreign firms, especially privately owned, differ in their political connections to their home government, dependence on state support, and the commercial incentives to preserve relationship with the host country (Pearson, Rithmire, and Tsai, 2022; Stone, Wang, and Yu, 2022). Accordingly, even though firms are not immune to pressures from their home governments, they can often resist them and refuse particular demands. For example, Microsoft, a U.S. firm operating a data centre in Dublin, challenged a 2013 U.S. warrant seeking the disclosure of customer email content stored in that data centre by invoking the privacy protections applicable to its foreign-held data.[5] The host government may not know ex-ante whether a foreign firm will cooperate with its home government or resist its demands, and can, therefore, remain uncertain about the firm's type.

Unverifiable Signal During Peace. We assume that the signal that the government receives during peace is private and unverifiable. In the context of investment screening,

[5]Microsoft challenged the order citing possible complications with other countries' domestic data privacy laws and explicitly stated "The power to embark on unilateral law enforcement incursions into a foreign sovereign country – directly or indirectly – has profound foreign policy consequences...". See https://www.theguardian.com/technology/2014/dec/14/privacy-is-not-dead-microsoft-lawyer-brad-smith-us-government.

the information that the government gets about latent capacity may originate as classified intelligence or other channels where the government cannot credibly demonstrate this to the firm or any other adjudicating authority. Such a setting where the principal has information that is non-contractible is common across many applications (Aghion, Dewatripont, and Rey, 1994; Baker, Gibbons, and Murphy, 1994; Fuchs, 2007; Levin, 2003). Thus, the peace-time contract only depends on the calendar time.[6] By contrast, the signal during hostility is generated by an act of undermining for which the government can produce verifiable evidence. Thus, we allow the government to condition the hostility contract on the history of detected undermining.

Pooling Contracts. We restrict the government to a single contract within each geopolitical state. A single pooling peace-time contract rules out a menu of contracts in which the firm's selection reveals its type during peace. This restriction captures the political cost that an adversarial firm may face from its home country if it voluntarily reveals itself to the host government. Publicly observed choices that reveal a player's alignment may have strategic consequences (Bils and Smith, 2025; Smith, 2021). In the context of investment screening, selecting a contract intended for an adversarial firm would constitute an observable act of cooperation with the host government which could result in the firm's home government imposing significant costs on the firm. In particular, foreign firms may depend on their home governments for financing, licenses, or even legal protection which allows the home government to discipline these firms (Davis, Fuchs, and Johnson, 2019; Li, Meng, et al., 2008; Pearson, Rithmire, and Tsai, 2022). The resulting political costs could be sufficiently large that an adversarial firm would not voluntarily choose a contract that reveals its type. These potential consequences for the adversarial firm motivate our restriction to a single pooling contract during peace.

We impose the same single-contract restriction during hostility. However, this restriction is without loss in the hostility game: the government cannot improve by offering a type-

[6] To clarify, the signal is unverifiable, so it cannot enter the contract whereas access schedules are published. So any differential access would be for a reason the government cannot state and hence, will invite challenges from other investors.

contingent menu in a zero-sum game and may restrict attention to a single stringent pooling contract.

3 Analysis

The government offers a peace-time contract $\mathcal{C}^p \in \mathfrak{C}^p$ at $t = 0$, where $\mathcal{C}^p$ states an exclusion time $T_R^p \in [0, \infty]$ and an exposure path $x^p = (x_t^p)_{t \leq T_R^p}$, with $x_t^p \in [\phi, 1]$, that applies when the relationship remains active during peace. If hostility arrives before the exclusion time, i.e., $T_h < T_R^p$, the peace-time contract expires at T_h. Although the contract allows for exclusion during peace, as the following lemma establishes, the government never prefers to do so.

Lemma 1. *Excluding the firm during peace is strictly suboptimal.*

Before hostility arrives at T_h, the payoffs remain aligned and the firm cannot impose security costs. If capacity building is detected, the government can always exclude the firm at the onset of hostility and prevent any loss due to undermining, since accumulated capacity can be used to inflict loss only while the firm retains access. For example, Norwegian authorities investigating security loopholes in Chinese-made electric buses found that removing SIM cards could prevent the manufacturer from remotely disabling these buses.[7] Hence, excluding the firm early only sacrifices the peace-time flow payoff and yields no additional security benefit. We can therefore restrict our attention to contracts with $T_R^p = \infty$, and define the peace time contract $\mathcal{C}^p$ only by its exposure path x^p.

After observing $\mathcal{C}^p$, the adversarial firm chooses its capacity-building effort, α_t, at each time during peace while the relationship remains active. Its effort can depend on the peace-time contract $\mathcal{C}^p$ as well as on its own history of capacity building choices, and may be randomized.

When hostility arrives at T_h, the government offers a contract $\mathcal{C}^h \in \mathfrak{C}^h$ specifying exposure and termination during hostility. Its choice of hostility contract can depend on its peace-time

[7] The potential security risks stemmed from the Chinese supplier's remote access to vehicles' control system that could be exploited. See https://www.theguardian.com/world/2025/nov/05/danish-authorities-in-rush-to-close-security-loophole-in-chinese-electric-buses for the investigation findings.

contract, $\mathcal{C}^p$, the hostility arrival date T_h, and its private detection during peace. Conditional on no peace-time detection, let q_{T_h} denote the government's posterior probability that the firm is commercial at the onset of hostility and P_{T_h} its posterior belief over capacity stock, C_{T_h}, at hostility if the firm is adversarial. During hostility, the firm chooses its undermining intensity β_t while the relationship continues. Its choice can depend on its entire history until t. The equilibrium concept is Perfect Bayesian Equilibrium.

4 Baseline: Persistent Geopolitical Risk

In the baseline, a geopolitical shock arrives randomly with hazard rate $\lambda > 0$. We solve the game backward, first characterizing the hostility continuation value for every belief inherited from peace and then solving the government's peace-time contracting problem using those continuation values.

Hostility Continuation

If capacity building was detected during peace, the government knows that the firm is adversarial and optimally excludes it when hostility begins. We therefore characterize the hostility continuation conditional on no peace-time detection. At T_h, the government has a posterior belief q_{T_h} that the firm is commercial and, conditional on it being adversarial, a posterior P_{T_h} over its capacity. For ease of notation, we write q for the posterior q_{T_h} at the end of peace and P for P_{T_h}. To characterize the hostility continuation, we first study a benchmark in which the adversarial firm's capacity accumulation is exogenously determined.

Benchmark: Exogenous Capacity Accumulation

Suppose the adversarial firm's capacity accumulates mechanically as a function of the peace-time exposure. Fix a peacetime exposure path x^p and suppose that the effort path α for the adversarial firm is exogenously fixed and known to the government. Then, the firm's

capacity stock at T_h is

$$C_{T_h}(x^p) = \int_0^{T_h} \alpha_t x_t^p \, dt = c.$$

Thus, at the onset of hostility, the government knows that the adversarial firm's capacity stock is c, so $P = \delta_c$. Under this effort path, the probability that the adversarial firm remains undetected throughout peace is $e^{-\mu \int_0^{T_h} \alpha_t x_t^p \, dt} = e^{-\mu c}$. Because a commercial firm never builds any capacity, it never generates a detection signal. Thus, if there is no detection during peace, the government's posterior belief that the firm is commercial at the onset of hostility is

$$q = q(c) = \frac{q_0}{q_0 + (1 - q_0)e^{-\mu c}}. \tag{1}$$

The expression for $q(c)$ reflects that detection is proportional to capacity building: accumulating capacity c leaves the adversarial firm undetected with probability $e^{-\mu c}$ regardless of when that capacity was acquired. Thus, for a given c, the timing of exposure and effort provides no additional information about the firm's type. Moreover, capacity remains fixed during hostility, and the government replaces the peace-time contract with a new hostility contract. Thus, conditional on no detection during peace, the exposure and effort paths affect the government's hostility contracting problem only through its posterior beliefs $q(c)$ and $P = \delta_c$. The latter pins down the adversarial firm's effective harm, $k(c) = 1 + Kc$. Given these beliefs, this contracting problem coincides with the screening problem in Kolb and Madsen (2023).

For a belief pair $(q(c), \delta_c)$ and a sufficiently small ϕ_h, the optimal hostility contract excludes the firm only upon detected undermining.[8] It grants full exposure during hostility if

$$q(c) \geq [1 - q(c)]Kc \quad \Longleftrightarrow \quad Kce^{-\mu c} \leq \frac{q_0}{1 - q_0}, \tag{2}$$

where the equivalence follows from (1) and $k(c) = 1 + Kc$. Note that full exposure follows

[8] They show it is without loss to use a stringent contract that excludes the firm upon the first detected undermining during hostility. A sufficiently small ϕ_h ensures that the optimal contract never excludes the firm in the absence of detection.

when the expected flow benefits, $q(c)$, are weakly greater than the expected flow loss from an adversarial firm undermining with full intensity, $[1-q(c)]Kc$. If instead, $q(c) \leq [1-q(c)]Kc$ at the start of hostilities, the optimal contract is a *testing contract*: exposure is initially restricted, then rises following continued non-detection, and eventually reaches one. The adversarial firm remains indifferent between undermining with full intensity, $\beta_t = 1$, and not undermining at all $\beta_t = 0$ whenever the exposure during hostility is $x_t^h < 1$. At $q(c) = [1-q(c)]Kc$, the two contracts coincide. Proposition A.6 characterizes the optimal hostility contract as well as the hostility payoffs.

The full exposure condition (2) combines the two opposing effects of capacity on the government's hostility contract through $Y(c) := Kce^{-\mu c}$: greater capacity increases the harm an adversarial firm can inflict, but accumulating that capacity also makes it less likely to escape detection during peace. The following proposition characterizes how capacity accumulated during peace affects the hostility contract.

Proposition 1 (Benchmark Hostility)**.** *Suppose the peace-time exposure path is x^p and effort α is exogenously fixed so that the capacity stock at the onset of hostility at T_h is c. Define $Y(c) := Kce^{-\mu c}$ and $C_Y = 1/\mu$. Then, conditional on no peace-time detection,*

1. *if $\frac{K}{\mu e} \leq \frac{q_0}{1-q_0}$, the government grants full exposure immediately for every capacity c.*
2. *if $\frac{K}{\mu e} \geq \frac{q_0}{1-q_0}$, there exist unique c^-, c^+, where $c^- \leq C_Y \leq c^+$, satisfying $Y(c^-) = Y(c^+) = \frac{q_0}{1-q_0}$, such that the government grants full exposure when $c \leq c^-$ or $c \geq c^+$, and offers a testing contract when $c \in [c^-, c^+]$.*

The two contracts coincide at $\frac{K}{\mu e} = \frac{q_0}{1-q_0}$ with $c^- = C_Y = c^+$. If the adversarial firm is detected before T_h, the government excludes the firm.

Proposition 1 implies that greater capacity need not lead to tighter restrictions during hostility. At low capacity, the potential harm is small; at high capacity, continued non-detection provides strong evidence that the firm is commercial. Testing can therefore be optimal only at intermediate capacities. Whether such a testing region exists depends on whether the

maximum value that $Y(c)$ attains, $Y(C_Y) = K/(\mu e)$, exceeds the prior odds $q_0/(1-q_0)$ that the firm is commercial.

Two inferences from the benchmark hostility are worth stating explicitly. First, the effort path affects the hostility contract only through the accumulated capacity. Thus, conditional on no detection during peace, the hostility contract that the government offers depends only on the capacity that it believes the firm accumulated given the peace-time exposure path; and the firm's own capacity-accumulation history enters its problem only through its capacity stock, C_t, since its probability of remaining undetected is $e^{-\mu C_t}$ along every path that reaches C_t. Second, greater capacity need not increase the adversarial firm's expected hostility payoff. It faces the same two effects: greater capacity implies more harm to inflict, but also a greater chance of being detected during peace. We next examine how the adversarial firm resolves this tradeoff, and how its choice shapes the government's beliefs and hostility contract.

Strategic Capacity Accumulation

Fix a peace-time exposure path x^p, and let $X_t = \int_0^t x_s^p \, ds$ denote the cumulative exposure through t. Conditional on relations remaining peaceful through t, the adversarial firm's capacity stock is $C_t = \int_0^t \alpha_s x_s^p \, ds$ along any effort path $\alpha = (\alpha_s)_{s \leq t}$ it chooses. Since $\alpha_s \in [0,1]$ at every instant, the adversarial firm can accumulate any capacity $C_t \in [0, X_t]$. The first inference from the benchmark holds irrespective of whether the effort α is fixed exogenously or chosen strategically, since the cumulative peace-time detection hazard is μC_t along every effort path that yields C_t. Thus, conditional on no detection, the hostility contract depends on the peace-time exposure path x^p only through the capacity that the adversarial firm accumulates, and that capacity is constrained by the exposure path only through X_t.

Because the government offers $\mathcal{C}^p$ at $t = 0$, the adversarial firm knows, when choosing its effort α_t, the complete exposure path x^p for as long as the relationship remains peaceful. Suppose hostility begins at T_h when the cumulative exposure is $X_{T_h} = X$. We first characterize what capacity the firm would optimally have accumulated by then and the hostility

contract the government would offer. We then show that the firm's effort choices can generate a single feasible capacity process $(C_t)_{t\geq 0}$ that attains these outcomes at every possible hostility date, even though the firm does not know T_h in advance.

Let H denote the distribution over capacity that the adversarial firm's effort choice induces on $[0, X]$. Because an adversarial firm that accumulates capacity c remains undetected until hostility with probability $e^{-\mu c}$, the government's posterior over capacity and its belief that the firm is commercial is

$$P(A) = \frac{\int_A e^{-\mu c}\ dH(c)}{\int e^{-\mu c}\ dH(c)}, \qquad q_{T_h} = \frac{q_0}{q_0 + (1-q_0)\int e^{-\mu c}\, dH(c)} \tag{3}$$

which reduce to δ_c and (1) when H is degenerate.

At the onset of hostility, the government offers a contract $\mathcal{C}_X^h$ that is optimal given its beliefs q_{T_h} and P. For this hostility contract, the adversarial firm's expected hostility payoff as a function of H is [9]

$$U^F(H, \mathcal{C}_X^h; X) = \int_0^X U^F(c; \mathcal{C}_X^h)\, dH(c) = \int_0^X e^{-\mu c} u^h(\mathcal{C}_X^h, c)\, dH(c), \tag{4}$$

where $u^h(\mathcal{C}_X^h, c)$ denotes the adversarial firm's continuation value under optimal undermining at the onset of hostility when accumulated capacity is c, and $U^F(c; \mathcal{C}_X^h) = e^{-\mu c} u^h(\mathcal{C}_X^h, c)$ is its expected hostility payoff for that c. An equilibrium at cumulative peace-time exposure X is a pair $(H_X^*, \mathcal{C}_X^{h*})$ in which the government's beliefs, given by (3), follow from H_X^*, the hostility contract $\mathcal{C}_X^{h*}$ is optimal given those beliefs, and H_X^* is optimal given that contract.[10]

To see why the equilibrium requires a distribution over capacities, suppose the government anticipates the adversarial firm accumulates capacity $\hat{c}$ and offers the corresponding benchmark hostility contract $\mathcal{C}_X^h(\hat{c})$.[11] For this to constitute an equilibrium, the firm must prefer $\hat{c}$

[9] The adversarial firm's peace flow payoff is determined by x^p alone and not by its effort. Therefore, the firm chooses capacity to maximize (4).

[10] The equilibrium concept is PBE here. The pair $(H_X^*, \mathcal{C}_X^{h*})$ denotes the capacity distribution and hostility contract in a PBE of the game with fixed cumulative exposure X.

[11] A degenerate belief $P = \delta_{\hat{c}}$ enters the hostility problem only through $k(\hat{c}) = 1 + K\hat{c}$, so the contract is

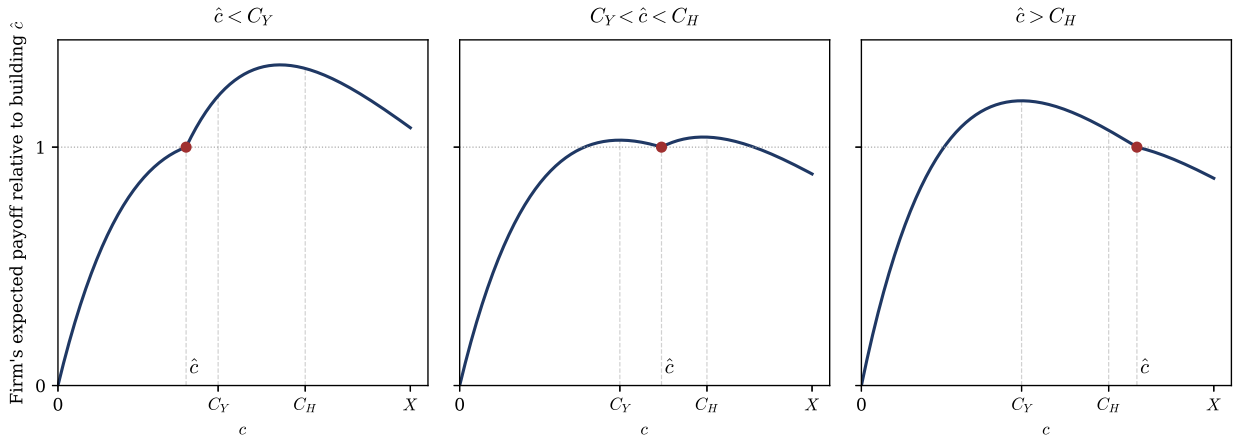


Figure 2: Deviations under a fixed hostility contract. The government offers $\mathcal{C}_X^h(\hat{c})$. Each panel plots the ratio $\frac{U^F(c;\mathcal{C}^h(\hat{c}))}{U^F(\hat{c};\mathcal{C}^h(\hat{c}))}$. Here, $C_H = \min\left\{c \geq C_Y : \mu q(c)\left(c + \frac{1}{K}\right) \geq 1\right\}$ and $c \in [0, X]$. Parameters: $q_0 = 0.15$, $K = 0.30$, $\mu = 0.20$, $r = 0.02$, $\gamma = 1$, and $X = 11$, with $\hat{c} = 4.0$, 6.3, 8.6, respectively.

to every other feasible capacity under that contract. As Figure 2 demonstrates, when $\mathcal{C}_X^h(\hat{c})$ is a testing contract (defined in Proposition A.6) and $X > C_Y$, the adversarial firm has a profitable deviation for any $\hat{c}$, provided the deviation is feasible, i.e., $c \leq X$.

Every capacity in the support of H_X^* must yield the same expected hostility payoff to the adversarial firm. A higher capacity implies a lower probability of the adversarial firm reaching hostility undetected. Thus, the hostility contract $\mathcal{C}_X^{h*}$ must offer a higher continuation value to an adversarial firm that has accumulated more.

The government excludes the firm at the onset of hostility if it detects capacity building at any time during peace. Let $V^h\left(\mathcal{C}_X^{h*}, H_X^*\right)$ denote its continuation value at the onset of hostility conditional on no peace-time detection. Its expected hostility payoff is therefore

$$\Psi(X) = \left[q_0 + (1 - q_0)\int_0^X e^{-\mu c}\, dH_X^*(c)\right] V^h\left(\mathcal{C}_X^{h*}, H_X^*\right), \tag{5}$$

where the term in the bracket is the probability of no detection before hostility. We characterize this equilibrium by applying the endogenous-harm screening analysis in Zalke (2026),

pinned down by the pair $(q(\hat{c}), k(\hat{c}))$ and we index it by $\hat{c}$ alone for notational ease.

accounting for the constraint that the firm's capacity cannot exceed cumulative peace-time exposure X.[12] The following proposition gives the equilibrium capacity distribution and the government's resulting hostility payoff.

Proposition 2 (Equilibrium at X)**.** *Suppose the cumulative peace-time exposure is fixed at $X \geq 0$. Define $\underline{c}_X = \min\{X, C_Y\}$ and $\bar{c}_X = \min\{X, C_H\}$, where*

$$w(c) = \mu q(c)\left(c + \frac{1}{K}\right), \quad C_H = \min\{c \geq C_Y : w(c) \geq 1\}, \quad M(c) = (1 + Kc)\big(1 - q(c)\big).$$

Then, in any equilibrium at X,

- *the capacity distribution is unique and has the following functional form:*

$$H_X^*(c) = \begin{cases} 0, & c < \underline{c}_X, \\ \dfrac{1 + \big(c - \underline{c}_X\big)K}{M\big(\bar{c}_X\big)}, & \underline{c}_X \leq c < \bar{c}_X, \\ 1, & c \geq \bar{c}_X. \end{cases} \tag{6}$$

- *The government's ex ante hostility payoff is*

$$\Psi(X) = \begin{cases} \dfrac{q_0}{r} - \dfrac{1 - q_0}{r + \gamma} K \underline{c}_X e^{-\mu \underline{c}_X}, & X \leq c^-, \\ \dfrac{q_0 \gamma}{r(r + \gamma)} e^{-rD(X)}, & X \geq c^-, \end{cases} \tag{7}$$

where c^- is as characterized in Proposition 1, with $c^- = \infty$ when $K/\mu e \leq q_0/(1 - q_0)$, and

$$D(X) = \frac{\mu}{2\gamma K}\Big[(1 + K\bar{c}_X)^2 - (1 + K\underline{c}_X)^2\Big] - \frac{1 + K\bar{c}_X}{\gamma} \log\left(\frac{w(\bar{c}_X)}{\mu \underline{c}_X}\right).$$

Proposition 2 shows how the firm's capacity choices respond to the peace-time contract $\mathcal{C}^p$ through the cumulative peace-time exposure X, as illustrated in Figure 3. When $X \leq C_Y$, where $C_Y = 1/\mu$, the adversarial firm chooses to accumulate X: the firm uses all available exposure to build capacity.[13] When $C_Y < X < C_H$, we can see from Figure 2 that the ad-

[12] In the endogenous-harm screening problem studied in Zalke (2026), a principal offers a single hostility contract against a distribution of privately known capacities that is selected endogenously by an agent. Unlike in the problem studied in Zalke (2026), here, the firm's capacity choice is restricted to $[0, X]$, where X is determined by the government's peace-time exposure policy.

[13] The adversarial firm's expected hostility payoff is proportional to $Y(c) = Kce^{-\mu c}$ for $c \leq C_Y$. Thus, it will accumulate as much as it can until C_Y.

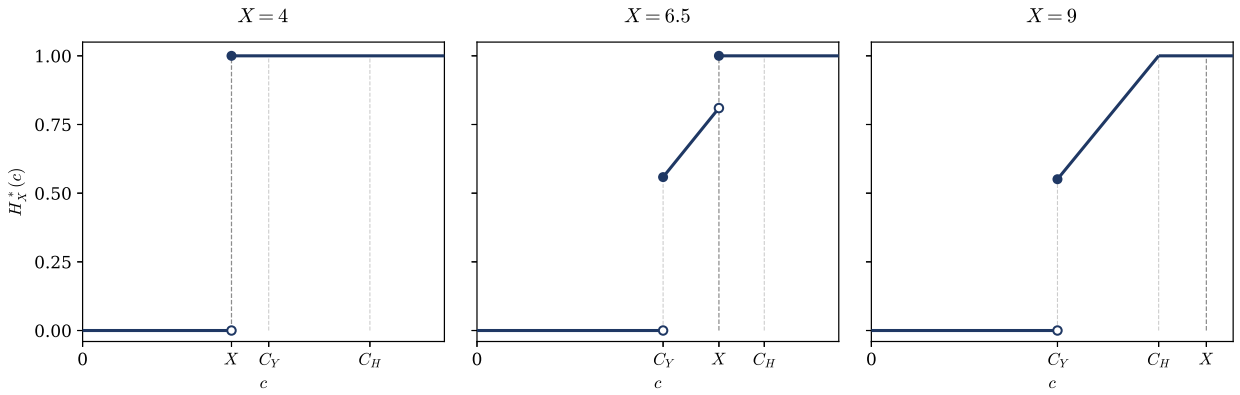


Figure 3: Equilibrium capacity distributions at different levels of cumulative peace-time exposure. Parameters: $q_0 = 0.15$, $K = 0.30$, and $\mu = 0.20$, implying $C_Y = 5$ and $C_H = 7.72$.

versarial firm has a profitable deviation to any hostility contract calibrated to $c \in (C_Y, C_H)$. Thus, no single capacity c can be an equilibrium. The equilibrium distribution instead has support $[C_Y, X]$, with atoms at both endpoints and a constant density in the interior. Additional exposure raises the upper bound of this distribution, while the probability mass at X declines. Beyond C_H, the additional gains from accumulating capacity are not sufficient to compensate for the increased detection risk. Therefore, once $X \geq C_H$, the firm does not accumulate any further capacity and the distribution remains unchanged.

Proposition 2 characterizes the equilibrium separately at any cumulative peace-time exposure X. The adversarial firm, however, chooses its effort sequentially without knowing when hostility will arrive. Therefore, for a peace-time contract $\mathcal{C}^p$ with the exposure path x^p, the firm's choice over effort must generate a feasible capacity process $(C_t)_{t\geq 0}$ that satisfies $C_t \sim H^*_{X_t}$ at every t, and not just for the one when hostility actually arrives. The following lemma shows that such a process exists.

Lemma 2 (Dynamic Implementation)**.** *Fix a peace-time contract $\mathcal{C}^p$ with exposure path x^p. There exists a feasible capacity process $(C_t^*)_{t\geq 0}$ such that $C_t^* \sim H^*_{X_t}$ for every t, where H^*_X is characterized in Proposition 2. Moreover, in any PBE, the firm's capacity following $\mathcal{C}^p$ satisfies $C_t^* \sim H^*_{X_t}$ at every t during peace at which $\lambda > 0$.*

It follows from Lemma 2 that the adversarial firm can attain its optimal expected hostility

payoff at every possible hostility arrival date under the corresponding hostility contracts. The lemma pins down the capacity distribution at each cumulative exposure. Any effort choice that induces this capacity process is optimal even though the firm does not know when hostility arrives. The government can consequently evaluate a peace-time exposure path, x^p, using $\Psi(X_t)$ (given in (7)), as its expected hostility payoff if hostility arrives at t.

Peace-time Exposure

During peace, exposure generates economic gains while also allowing the adversarial firm to accumulate more capacity, which affects the government's payoff when hostility begins. If hostility begins at t, Proposition 2 and Lemma 2 imply that the government's expected hostility payoff is $\Psi(X_t)$. Thus, if hostility arrives at T_h, the government's expected payoff from exposure x^p is

$$\int_0^{T_h} e^{-rt} x_t^p \, dt + e^{-rT_h} \Psi(X_{T_h}),$$

where the first term captures the economic gains received during peace, and the second term is the discounted expected hostility payoff. Since hostility arrives at rate $\lambda > 0$, the government's peace-time problem is to choose an exposure path x^p to maximize

$$W(x^p) = \int_0^\infty e^{-(r+\lambda)t} \left[x_t^p + \lambda \Psi(X_t)\right] dt, \tag{8}$$
$$\text{s.t. } \dot{X}_t = x_t^p, \quad X_0 = 0, \quad x_t^p \in [\phi, 1].$$

Because the government's expected hostility payoff Ψ depends on the exposure path only through the cumulative exposure X_t at that time, and the hazard λ is constant, the government's continuation value, $V(X)$, for any cumulative exposure X, satisfies the following Hamilton–Jacobi–Bellman (HJB) equation

$$(r+\lambda)V(X) = \max_{x \in [\phi,1]} \left\{x + \lambda \Psi(X) + xV'(X)\right\}. \tag{9}$$

The maximand is linear in x, so the optimal peace-time exposure path sets $x_t^p = 1$ when

$1 + V'(X_t) > 0$, and minimal exposure, $x_t^p = \phi$, when $1 + V'(X_t) < 0$. Any $x_t^p \in [\phi, 1]$ is optimal when $1 + V'(X_t) = 0$.[14] Intuitively, increasing current exposure, x_t^p, raises the government's flow payoff as well as the rate at which the cumulative exposure X_t grows. A marginal increase in x_t^p therefore offers an economic gain of one but incurs a marginal loss of $-V'(X_t)$ in continuation value. For a given cumulative exposure X_t, neither the marginal economic gain nor the marginal loss depends on the current access level x_t^p, so if one unit of exposure is worthwhile, so are all of them.

To determine when the government switches between minimum and full access, consider how cumulative exposure affects its expected hostility payoff. Define

$$G(X) = -\Psi'(X) \tag{10}$$

as the marginal security cost of cumulative exposure X. It is the marginal loss in the government's expected hostility payoff from additional cumulative exposure. This cost reflects both the firm's response to additional opportunities to accumulate capacity and the resulting hostility contract. From Proposition 2, when $X \geq C_H$, additional cumulative exposure changes neither the equilibrium capacity distribution nor the government's expected hostility payoff. The marginal security cost is therefore zero. The following lemma establishes that this cost decreases strictly as cumulative exposure approaches C_H.

Lemma 3 (Marginal security cost)**.** *$G(X)$ is continuous on $[0, \infty)$, strictly decreasing on $[0, C_H)$, and equal to zero for all $X \in [C_H, \infty)$, where C_H is defined in Proposition 2.*

Figure 4 illustrates the declining marginal security cost. As the figure shows, marginal cost remains strictly positive for $X < C_H$. Thus, increasing cumulative exposure lowers the government's expected hostility payoff, $\Psi(X)$, even as the marginal security cost falls.

Since the marginal security cost is zero for $X \geq C_H$ (Lemma 3), the government must grant full access, $x_t^p = 1$, whenever the cumulative exposure is weakly greater than C_H. To see

[14] When the government is indifferent between two paths, we select a right-continuous exposure path. This tie-breaking convention does not affect equilibrium payoffs or the exposure path a.e.

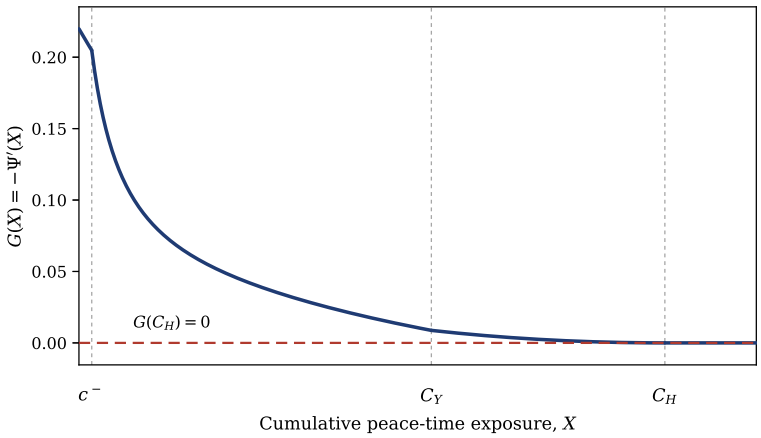


Figure 4: Government's marginal security cost of cumulative exposure. Parameters: $r = 0.02$, $\gamma = 1.00$, $q_0 = 0.15$, $K = 0.30$, and $\mu = 0.20$.

how much earlier full access becomes optimal, consider a cumulative exposure $X \leq C_H$ from which point on the government grants full access. Under this policy, the government receives a flow payoff of one, and cumulative exposure reaches $X + t$ after another t units of peace. Its continuation value is therefore

$$V(X) = \int_0^\infty e^{-(r+\lambda)t} \left[1 + \lambda \Psi(X + t)\right] dt.$$

Increasing X raises the cumulative exposure at every future date by the same amount, while leaving the flow payoff unchanged. Thus, the marginal loss in continuation value is

$$-V'(X) = \lambda \int_0^\infty e^{-(r+\lambda)t} G(X + t)\, dt = \lambda \int_X^{C_H} e^{-(r+\lambda)(s-X)} G(s)\, ds, \tag{11}$$

where the second equality uses a change of variable, $s = X + t$, and the fact that $G(s) = 0$ for $s \geq C_H$ (Lemma 3).

As equation (11) shows, the marginal loss in continuation value aggregates the marginal security costs at possible future hostility dates. As X increases, the marginal security cost $G(X + t)$ decreases at each subsequent time t. The marginal loss in continuation value under full access therefore decreases with X and reaches zero at C_H. When the government

switches from offering minimum exposure to maximum, this loss $-V'(X)$ equals the marginal economic gain from increasing current access, x_t^p, which is one. At higher cumulative exposure, the marginal security costs are smaller, making full access optimal thereafter. The following proposition characterizes the resulting exposure path.

Proposition 3 (Persistent Geopolitical Risk)**.** *Fix a $\lambda > 0$. The government's optimal peace-time exposure path can have at most one switch, $\tilde{\tau}^*$ such that*

$$x_t^{p*} = \begin{cases} \phi, & t < \tilde{\tau}^*, \\ 1, & t \geq \tilde{\tau}^*, \end{cases}$$

where $\tilde{\tau}^* = 0$ if $\lambda \int_0^{C_H} e^{-(r+\lambda)(s)} G(s)\, ds \leq 1$. Otherwise, $\widetilde{\tau}^* \in (0, C_H/\phi)$ is the unique solution to

$$1 = \lambda \int_{\phi\widetilde{\tau}^*}^{C_H} e^{-(r+\lambda)(s-\phi\widetilde{\tau}^*)} G(s)\, ds,$$

where G is the marginal security cost defined in equation (10).

Proposition 3 shows the probationary pattern in the government's exposure path: it may initially restrict exposure and later liberalize. Figure 5 illustrates this single-switch exposure path: at $\widetilde{\tau}^*$, the government switches from offering minimal exposure, $x_t^{p*} = \phi$, to maximum exposure, $x_t^{p*} = 1$. Since exposure is held constant at ϕ through the probationary period, cumulative exposure at the switch is $\widetilde{X} = \phi\tilde{\tau}^*$. Since $\tilde{\tau}^* < C_H/\phi$, the cumulative exposure at the end of probation, $\widetilde{X}$, is less than C_H, which is the cumulative exposure at which the marginal security cost equals 0. The marginal security cost is therefore positive at the end of the probationary period, so the government liberalizes even though additional exposure still lowers its expected hostility payoff.

This result highlights the interaction between the three effects of exposure: the economic gain, the potential harm due to an adversarial firm's capacity accumulation, and the information generated due to monitoring. The harm and monitoring effects jointly determine the marginal security cost, $G(X_t)$. It is easiest to see this for the case $K/(\mu e) \leq q_0/(1-q_0)$, in which the government grants full exposure immediately when hostility begins. The hostil-

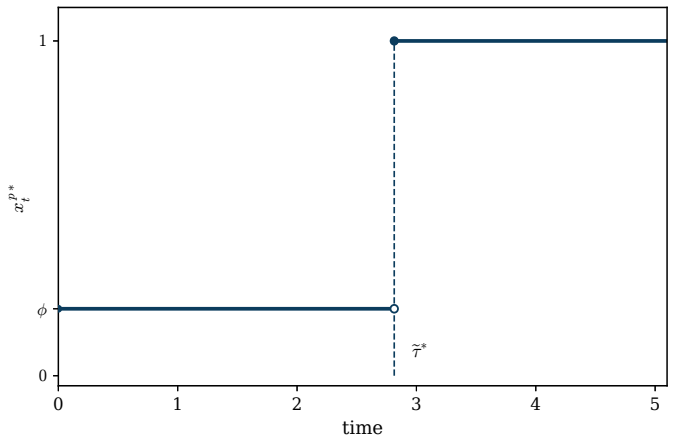


Figure 5: Peace-time stakes curve x_t when hostility arrival is random. Parameters: $r = 0.01$, $\gamma = 0.05$, $\lambda = 0.50$, $\phi = 0.20$, $q_0 = 0.70$, $K = 1.11$, and $\mu = 0.40$. The switching time is $\widetilde{\tau}^* = 2.814$.

ity contract then remains unchanged as cumulative exposure increases. For $X_t \leq C_Y$, the adversarial firm uses all available exposure to accumulate capacity. From (7), the marginal security cost is

$$G(X_t) = -\Psi'(X_t) = \underbrace{\frac{(1-q_0)K}{r+\gamma} e^{-\mu X_t}}_{\text{harm effect}} - \underbrace{\frac{(1-q_0)K}{r+\gamma} \mu X_t e^{-\mu X_t}}_{\text{monitoring effect}}. \tag{12}$$

The harm effect captures the additional cost from greater capacity when the adversarial firm remains undetected. The monitoring effect captures the cost avoided because additional capacity accumulation increases the probability of detection and subsequent exclusion.

Because the marginal security cost is non-negative (Lemma 3), it follows that the harm effect always (weakly) dominates the monitoring effect. In fact, because $\widetilde{X} < C_Y$,[15] the harm effect strictly dominates the monitoring effect when the government liberalizes. However, as the cumulative exposure increases, the harm effect weakens while the monitoring effect strengthens so that the gap between the two narrows making additional cumulative exposure less costly. Eventually, these lower marginal security costs reduce the marginal loss in

[15]From Proposition 3, $\widetilde{X} < C_H$ but by the definition of C_H, it is equal to C_Y when $K/(\mu e) \leq q_0/(1-q_0)$.

continuation value enough for the economic gains from greater current exposure to justify liberalization.

Discussion

India's regulation of investments, since the year 2020, from countries sharing a land border with it provides a useful case for examining screening under persistent geopolitical risk. The India-China relationship has historically been tense given their border dispute (Holslag, 2009; Ogden, 2022). The period following 2019 was especially high risk given the border skirmish that year.[16] Even as economic engagement between the two countries continued, there was a persistent risk of military escalation that would mark the arrival of hostility. Thus, this makes the case relevant to our setup, where geopolitical risk is persistent.

In April 2020, the Indian government announced that foreign direct investment (FDI) from neighboring countries would require prior government approval.[17] Although the policy covered all countries that share a land border with India, contemporary reports identified Chinese investments as a particular concern. Officials in the Indian government had expressed concerns about growing Chinese investments in strategically sensitive sectors, with one official articulating their concern as "We have been particularly suspicious of investments from China in sectors such as telecommunications and power as they are of strategic importance."[18]

These restrictions continued until 2026, when the Indian government amended this earlier policy to ease some of the restrictions. It allowed investors with up to 10 percent ownership from land-bordering countries to invest without needing government approval.[19] The government justified these changes by emphasizing "access to new technologies" and the op-

[16] "Indian, Chinese Soldiers Get into Scuffle in Ladakh," https://timesofindia.indiatimes.com/india/indian-chinese-soldiers-get-into-scuffle-in-ladakh/articleshow/71088887.cms.

[17] See https://www.pib.gov.in/PressReleasePage.aspx?PRID=1615711 for changes in India's Foreign Direct Investment policy.

[18] "China Investments Come Under Greater Scrutiny," https://www.livemint.com/news/india/china-investments-come-under-greater-scrutiny-11592502650311.html.

[19] See https://www.pib.gov.in/PressReleasePage.aspx?PRID=2301992 for the amended policy.

portunity to "supplement domestic capital," highlighting the economic motivations behind the easing.[20] Importantly, this easing occurred even as geopolitical tensions persisted. In an interview in 2026, India's Army Chief emphasized the need to maintain a strong military deployment along the India-China border.[21] Thus, India moved from broad restrictions, requiring approval for every inward investment from China, to selective liberalization while retaining oversight and continuing to prepare for possible military escalation.

This evolution in regulatory policy illustrates a restriction followed by some liberalization, even as monitoring continues and geopolitical risk persists. Proposition 3 offers an insight into the mechanism through which the policy can take such a shape. As the results show, conditional on peace continuing and non-detection, the interaction between the information generated through monitoring and its effect on an adversarial firm's choice of capacity stock reduces the marginal security cost of exposure sufficiently. Once this cost falls sufficiently, the economic gains justify liberalization. Greater access can, therefore, be optimal even while uncertainty about the firm remains and its home country continues to be a potential adversary.

5 Anticipated Geopolitical Risk

The preceding analysis assumes that the risk of hostility is present from the beginning of the relationship. We now consider an anticipated increase in geopolitical risk. Hostility remains stochastic: it cannot arrive before a publicly known date T and, conditional on not yet having arrived, has a constant hazard $\lambda > 0$ thereafter. Formally, the hazard rate is

$$\lambda(t) = \begin{cases} 0, & t < T, \\ \lambda, & t \geq T. \end{cases}$$

[20] The government's press release highlights access to capital and technology to support domestic manufacturing. See https://www.pib.gov.in/PressReleasePage.aspx?PRID=2237806.

[21] General Upendra Dwivedi referred to the situation as "Stable, but constant vigil...". See https://www.newindianexpress.com/india/2026/Jun/09/interview-army-gearing-up-for-a-two-and-a-half-front-challenge.

Thus, T marks the onset of the risk of hostility rather than the realization of hostility itself.

Let $X_T = \int_0^T x_s^p \, ds$ denote the cumulative exposure when geopolitical risk begins. Because the hostility hazard is persistent from T onward, Lemma 2 implies that the equilibrium capacity distribution satisfies $H_t = H^*_{X_t}$ for every $t \geq T$. From T onward, the government faces the persistent-risk problem characterized in the preceding section, starting from state X_T with continuation value $V(X_T)$. Thus, for any feasible level of cumulative exposure X_T, the following lemma characterizes its optimal allocation over the pre-T period.

Lemma 4 (Front-loaded Exposure)**.** *Fix $X \in [\phi T, T]$. Among all admissible exposure paths x^p satisfying $X_T = \int_0^T x_t^p \, dt = X$, the government's discounted peace payoff $\int_0^T e^{-rt} x_t^p \, dt$ is maximized by the bang–bang (front-loaded) path*

$$x_t^B(X) \;=\; \begin{cases} 1, & 0 \leq t < \tau(X), \\ \phi, & \tau(X) \leq t < T, \end{cases}$$

where $\tau(X) \;=\; \dfrac{X - \phi T}{1 - \phi} \in [0, T]$.

The logic behind the lemma follows directly from discounting. Holding X_T fixed, shifting exposure from a later time to an earlier one changes neither the equilibrium capacity distribution nor the information inherited at T, and therefore leaves the continuation value $V(X)$ unchanged. It does, however, increase the government's discounted pre-T payoff. This yields the front-loaded path in Lemma 4.

Lemma 4 reduces the government's pre-T problem to choosing cumulative exposure $X \in [\phi T, T]$. Under the front-loaded path, its payoff is

$$W_A(X, T) = \frac{1 - e^{-r\tau(X)}}{r} + \phi \frac{e^{-r\tau(X)} - e^{-rT}}{r} + e^{-rT} V(X), \tag{13}$$

where $\tau(X) = \frac{X - \phi T}{1 - \phi}$. The first term is the peace-time benefit from full exposure before $\tau(X)$, the second is the benefit from minimal exposure between $\tau(X)$ and T, and the final term is the discounted value of entering the persistent-risk period with cumulative exposure X.

To characterize the optimal exposure before T, consider the effect of an additional unit of cumulative exposure X. By Lemma 4, this additional exposure extends the initial full-exposure phase and produces a marginal benefit of $e^{-r\tau(X)}$ before the risks begin. It also changes the discounted continuation value at T by $e^{-rT}V'(X)$. Thus, when the optimal choice of cumulative exposure before T lies between full restriction, ϕT, and full exposure, T, the two effects must balance:

$$-V'(X) = e^{r[T-\tau(X)]}. \tag{14}$$

Here, $-V'(X)$ is the marginal loss in continuation value of entering the risky period at T with an additional unit of exposure, while the right-hand side in equation (14) is the marginal peace-time benefit of exposure expressed in time-T units. For the remainder of the analysis, we maintain the following assumption on the minimum feasible exposure during peace.

Assumption 1. *We assume that the minimum exposure during peace, ϕ, is sufficiently small. Specifically,*

$$0 < \phi \;\leq\; \min\left\{1 - \frac{q_0\gamma}{r+\gamma},\;\; 1 - \frac{r}{\lambda\big[G(\widetilde{X}) - 1\big]}\right\},$$

where $\widetilde{X} = \phi\widetilde{\tau}^*$ and $\widetilde{\tau}^*$ is defined in Proposition 3.

Substantively, this assumption implies that the minimum feasible exposure that the government can restrict the foreign firms to is sufficiently small. Technically, for any fixed T, the assumption implies that the slope of $W_A(X,T)$ can cross from positive to negative at most once as X increases. Hence, $W_A(X,T)$ has at most one interior local maximum, denoted by $X^I(T) \in [\phi T, T]$,[22] and any global optimum must belong to the set $\{\phi T, X^I(T), T\}$.

Thus, for a given T, the government compares between three exposure paths for the pre-T period: minimum exposure before T, full exposure before T, and initial openness followed by restriction. We denote these choices by L, F, and I, respectively, and define the thresholds

[22] We allow $X^I(T)$ to coincide with either endpoint.

at which their payoff rankings change:

$$
\begin{aligned}
T_{IL}^{A} &:= \inf\left\{T > 0 : W_A\big(X^I(T), T\big) - W_A(\phi T, T) \geq 0\right\}, \\
T_{FL}^{A} &:= \inf\left\{T > 0 : W_A(T, T) - W_A(\phi T, T) \geq 0\right\}, \\
T_{FI}^{A} &:= \inf\left\{T > 0 : W_A(T, T) - W_A\big(X^I(T), T\big) \geq 0\right\},
\end{aligned}
\tag{15}
$$

where the definitions involving $X^I(T)$ apply only when it is feasible, i.e., $X^I(T) \in [\phi T, T]$. As we show in the appendix, these pairwise payoff differences are single-crossing in T, yielding the threshold characterization in the following proposition.

Proposition 4 (Anticipated Geopolitical Risk)**.** *Suppose that the hostility hazard is zero before T and equals a constant $\lambda > 0$ thereafter. Define $X_T(\tau) = \phi T + (1-\phi)\tau$, and $\widetilde{X} = \phi\widetilde{\tau}^*$, where $\widetilde{\tau}^*$ is defined in Proposition 3. Then, under Assumption 1, the government's optimal peace-time exposure path is*

$$
x_t^{p*} = \begin{cases} 1, & t < \tau_1^*, \\ \phi, & \tau_1^* \leq t < \tau_2^*, \\ 1, & t \geq \tau_2^*, \end{cases}.
$$

where,

- *the two switching times $0 \leq \tau_1^* \leq T \leq \tau_2^*$ are*

$$
\tau_1^* = \begin{cases} 0, & T < \underline{T}^A, \\ \tau^I(T), & \underline{T}^A \leq T < \overline{T}^A, \\ T, & T \geq \overline{T}^A, \end{cases}
$$

  and $\tau_2^* = T + \frac{1}{\phi}\max\left\{0, \widetilde{X} - X_T(\tau_1^*)\right\}$, respectively.

- *The thresholds are $\underline{T}^A = \min\{T_{IL}^A, T_{FL}^A\}$, and $\overline{T}^A = \max\{T_{FL}^A, T_{FI}^A\}$, where T_{IL}^A, T_{FL}^A, and T_{FI}^A are defined in equation (15). If $T_{FL}^A \leq T_{IL}^A$, $\tau^I(T)$ does not exist and $\underline{T}^A = \overline{T}^A = T_{FL}^A$.*

- *If $T_{FL}^A > T_{IL}^A$, $\tau^I(T) \in [0, T)$ is the unique optimal interior cutoff and satisfies*

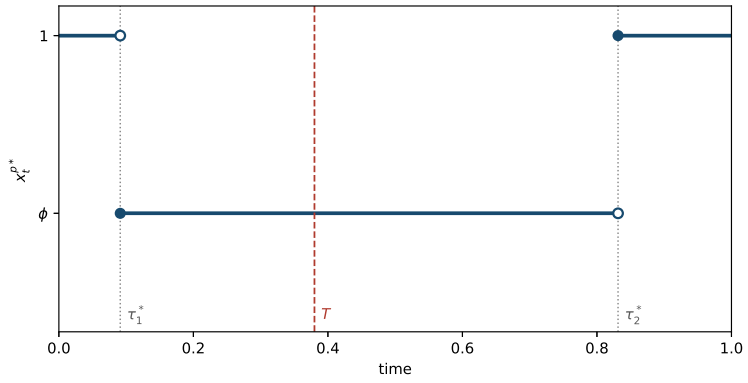


Figure 6: Peace-time stakes curve x_t^p under anticipated geopolitical risk. The hostility hazard is zero before $T = 0.38$ and equals $\lambda = 5.00$ thereafter. Parameters: $r = 0.10$, $\gamma = 0.01$, $\phi = 0.40$, $q_0 = 0.90$, $K = 1.19$, and $\mu = 0.05$. The government switches from full to minimal stakes at $\tau_1^* = 0.083$ and returns to full stakes at $\tau_2^* = 0.848$.

$$e^{-\frac{r+\lambda}{\phi}\left[\widetilde{X}-X_T(\tau^I(T))\right]} + \frac{\lambda}{\phi}\int_{X_T(\tau^I(T))}^{\widetilde{X}} e^{-\frac{r+\lambda}{\phi}\left[s-X_T(\tau^I(T))\right]} G(s)\,ds = e^{r\left(T-\tau^I(T)\right)},$$

where $G(\cdot)$ is defined in equation (10). The interior exposure is $X^I(T) = X_T(\tau^I(T))$.

Setting $T = 0$ in Proposition 4 recovers the persistent risk result characterized in Proposition 3. Because $0 \le \tau_1^* \le T$, setting $T = 0$ implies $\tau_1^* = 0$ and $X_T(\tau_1^*) = \phi T + (1-\phi)\tau_1^* = 0$. The second switching time then becomes $\tau_2^* = \widetilde{X}/\phi = \widetilde{\tau}^*$. Hence, the government grants minimal exposure until $\widetilde{\tau}^*$ and full exposure thereafter, exactly as characterized in Proposition 3.

Figure 6 illustrates the U-shaped exposure path that arises when $T > 0$ and the government chooses to enter the risky period with an interior level of cumulative exposure, i.e., $X^I(T) \in (\phi T, T)$. By Lemma 4, the government implements this choice by granting full exposure first and minimum exposure for the remainder of the pre-T period. The first switch is, therefore,

$$\tau_1^* = \frac{X^I(T) - \phi T}{1-\phi}.$$

Initial openness reflects the government's willingness to enter the risky period with greater cumulative exposure in exchange for economic gains before geopolitical risk emerges. Choosing

$X^I(T) > \phi T$ lowers its continuation value at T relative to maintaining minimum exposure, but provides additional economic gains earlier.

Restrictions continue after T because, from (14), the marginal loss in continuation value at the chosen exposure satisfies

$$-V'(X^I(T)) = e^{r[T-\tau_1^*]} > 1.$$

The economic gains earned before T justify accepting this marginal loss when choosing $X^I(T)$. However, when the geopolitical risk begins, $\lambda > 0$, the unit marginal economic gain of current exposure x_t^p is lower than the marginal loss in value, $-V'(X^I(T)) = e^{r[T-\tau_1^*]} > 1$. The government therefore maintains minimum exposure until, as Proposition 3 states, cumulative exposure reaches the persistent-risk reopening threshold, $\widetilde{X}$, where $-V'(\widetilde{X}) = 1$. Conditional on peace continuing, the government reopens at

$$\tau_2^* = T + \frac{\widetilde{X} - X^I(T)}{\phi}.$$

This results in the U-shaped ($1 \to \phi \to 1$) exposure path, as can be seen in Figure 6, where the government integrates initially, then restricts access before finally liberalizing.

Thus, Proposition 4 reveals an interesting reversal relative to the persistent-risk case in Proposition 3. Under persistent risk, the government initially restricts exposure and subsequently liberalizes; when risk is anticipated, it may instead grant full exposure, retrench, and then liberalize again. For a fixed X_T, front-loading changes neither the capacity accumulated nor the information acquired until T, and therefore, the expected marginal loss in continuation value at T, $-V'(X_T)$, remains unchanged. Instead, it raises the economic effect by delivering the economic gains from raising exposure earlier while hostility cannot occur.

The switching dates in the optimal exposure path in Proposition 4 depend on the duration of the anticipated period of peace. A longer period of peace implies greater delay between the economic gains of exposure and their potential security costs. This makes initial openness more attractive which affects both when restrictions begin as well as when reopening becomes

optimal.

Proposition 5 (Timing of Restriction)**.** *Suppose Assumption 1 holds. A longer anticipated period of peace weakly delays the onset of restrictions and weakly advances reopening whenever the optimal exposure path includes a period of restriction.*

As the duration of peace before the risky period increases, the marginal loss in continuation value is discounted more heavily due to the factor e^{-rT}, making initial openness more attractive and delaying the time at which the government restricts access, τ_1^*. Additionally, greater initial openness raises the cumulative exposure X_t at subsequent dates which lowers the marginal loss in continuation value, $-V'(X_t)$. The reopening condition, $-V'(X_t) = 1$, is therefore reached sooner bringing τ_2^* forward.

As we show in the discussion following Proposition 4, setting $T = 0$ recovers the persistent-risk setup. The comparative static in Proposition 5 also yields a useful comparison between the persistent-risk and anticipated-risk environment. Any initial full-exposure phase under anticipated risk moves cumulative exposure towards $\widetilde{X}$ at the maximum rate, shortening the subsequent retrenchment phase relative to persistent risk.

Corollary 1 (Restriction Comparison)**.** *Retrenchment is weakly shorter under anticipated geopolitical risk than under persistent geopolitical risk.*

Under persistent risk, the government offers the cumulative exposure up to $\widetilde{X}$ at the restricted rate ϕ. Under anticipated risk, it reaches $\tau_1^* > 0$ at the maximum rate before retrenchment begins, leaving only the interval from τ_1^* to $\widetilde{X}$ to be offered at the rate ϕ. Anticipation therefore delays the onset of restriction relative to persistent risk while also bringing liberalization forward, making the total retrenchment period shorter.

Discussion

U.S. policy towards Chinese investments is a useful case study of how investment screening evolves when policy-makers anticipate geopolitical risk in the future. In December 2004, the

National Intelligence Council highlighted China's growing military capabilities and identified the possibility of increasingly antagonistic U.S.-China relations by the year 2020 (National Intelligence Council, 2004, p. 55).[23] Thus, U.S. policymakers anticipated increasing tensions with China by a certain date in the future.

Despite this forecast, the U.S. investment policy towards China remained broadly open while simultaneously recognizing the security concerns. Speaking in Beijing in 2011, Treasury Assistant Secretary Marisa Lago highlighted Chinese firms' contribution to the U.S. economy while acknowledging the existence of these concerns.[24] Economic benefits, thus, formed part of the government's stated justification for maintaining openness despite recognized security concerns.

By June 2018, however, the White House was highlighting the threats that state-directed investment in critical technologies posed to national security, and calling for tighter restrictions on foreign investments. [25] Subsequently, the Foreign Investment Risk Review Modernization Act of 2018 (FIRRMA), enacted that August, expanded the scope of regulatory authority of the US regulatory body, the Committee on Foreign Investment in the United States (CFIUS), to scrutinize inward investments. A pilot program, effective from November 2018, extended CFIUS review to certain non-controlling investments in U.S. businesses involved in critical technologies, and introduced mandatory filings for transactions covered by the program (U.S. Department of the Treasury, 2018).

The sequence of U.S. investment policy features continued openness even after the identification of future geopolitical risks, followed by tighter restrictions before the forecast horizon. Proposition 4 characterizes how the timing of anticipated risk affects the ordering of openness and restriction. In the model, a government front-loads exposure as it wants the economic

[23]The report notes "...fears in the US of China as an emerging strategic competitor could fuel an increasingly antagonistic relationship...". See https://www.dni.gov/files/documents/Global%20Trends_Mapping%20the%20Global%20Future%202020%20Project.pdf.

[24]See https://home.treasury.gov/tg1366 for the full statement.

[25]The White House, "Statement from the President Regarding Investment Restrictions," June 27, 2018, https://trumpwhitehouse.archives.gov/briefings-statements/statement-president-regarding-investment-restrictions/.

gains sooner. Conditional on holding the cumulative exposure fixed at the beginning of risk, this front-loading has no effect on the firm's choice of capacity and the information available to the government when risk begins. This difference can generate an initial period of openness followed by restrictions as the risky period approaches. The results in this model also identify a subsequent phase: if peace continues without detection, continued monitoring and the firm's response can eventually make liberalization optimal again. This further reopening is an implication of the model. Applied to this case, it suggests that restrictions on Chinese investment could eventually be eased under continued oversight, even while China remains a potential adversary.

6 Conclusion

This paper studies how a government should regulate foreign firms' access over time under geopolitical uncertainty. Granting access to foreign firms generates economic benefits for the host country, but also allows an adversarial firm to develop capabilities that it can use to undermine the host country when relations between the host country and the firm's home country turn hostile. At the same time, developing these capabilities can expose an adversarial firm to the government. The firm therefore weighs the benefits of greater capabilities against the possibility of being identified, and the government must account for this response when choosing its exposure. Thus, economic gains, future security costs, and information about the firm's intentions are linked through both the government's and firm's strategic choices.

We show that when geopolitical risk begins only after an anticipated period of peace, the optimal exposure path can be U-shaped: the government initially grants full access, restricts it as the risky period approaches, and liberalizes again if peace continues. This runs counter to the logic of starting small as the government initially grants full access even though doing so allows the firm to develop greater capabilities for future hostility. Initial openness reflects a willingness to accept greater future security costs in exchange for economic gains before

geopolitical risk emerges.

By contrast, when geopolitical risk is persistent, the government may optimally start small, restricting access initially and liberalizing if peace continues. Such a probationary pattern is consistent with the starting small logic demonstrated in models of long-term relationships. In both environments, the government eventually liberalizes when the marginal loss of additional exposure falls below its economic gains. However, even at that point, additional cumulative exposure lowers the government's expected hostility payoff. Therefore, liberalization occurs even though geopolitical risk persists and, conditional on no detection, the firm's intentions remain uncertain.

These results emphasize how the same economic interests, security concerns, and uncertainty about firm's intentions can produce different sequences of openness and restriction depending upon when geopolitical risk begins. Following extensions provide directions for future work:

- Verifiable Peace-time Signal: Peace-time detection could generate verifiable evidence which would allow the government to condition its peace-time contract on this detection. This would expand the government's contracting choice.
- Hostility Conditional on Actions: The arrival rate of hostility could depend on the capacity accumulated by the firm. Exposure would then affect not only the costs that an adversarial firm can impose during hostility but also when hostility occurs. In this setup, continued peace itself becomes informative about the hidden capacity.

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A Screening Contract Under Hostility

The following proposition describes the contract during hostility as well as the continuation payoffs

Proposition A.6 (From Theorem 1, Kolb and Madsen (2023)). *Suppose hostility begins at T_h, no capacity building detected during peace, the adversarial firm's capacity stock is c, and the government's beliefs over capacity is $P^C = \delta_c$. Let $t \geq 0$ denote the time elapsed since the onset of hostility. Given posterior $q(c)$ and effective harm $k(c)$, the optimal hostility contract, $\mathcal{C}^{h*}(q(c), k(c))$, is a stringent contract (termination upon detection of undermining) that is pinned down by $q(c)$ and $k(c)$ such that,*

(i) High Posteriors: *If $q(c) \geq \frac{k(c)-1}{k(c)}$, the government sets $x_t = 1$ for all $t \geq 0$ and the adversarial firm always undermines with full intensity, $\beta_t = 1$ for all $t \geq 0$. The adversarial firm and government's hostility continuation payoffs are, respectively,*

$$u^h(\mathcal{C}^{h*}(q(c), k(c)), c) = \frac{k(c)-1}{r+\gamma} \quad ; \quad V^h(\mathcal{C}^{h*}(q(c), k(c)), c) = \frac{q(c)}{r} - (1-q(c))\left(\frac{k(c)-1}{r+\gamma}\right)$$

(ii) Low Posteriors: *If $q(c) \leq \frac{k(c)-1}{k(c)}$, the government offers a testing contract: the unique optimal stakes path begins immediately in an escalation phase with the stakes growing continuously at rate $r + \frac{\gamma}{k(c)}$ followed by a discrete jump to $x_t = 1$ at time $\bar{t}_l = \frac{k(c)}{\gamma} \log\left(\frac{k(c)-1}{k(c)q(c)}\right)$. The adversarial firm is indifferent between undermining with full intensity, $\beta_t = 1$, and not undermining at all, $\beta_t = 0$ for all $t \geq 0$. The adversarial firm and government's hostility continuation payoffs are, respectively,*

$$u^h(\mathcal{C}^{h*}(q(c), k(c)), c) = \frac{k(c)-1}{r+\gamma}\left(\frac{q(c)k(c)}{k(c)-1}\right)^{1+\frac{rk(c)}{\gamma}} \quad ; \quad V^h(\mathcal{C}^{h*}(q(c), k(c)), c) = \frac{q(c)\gamma}{r(r+\gamma)}\left(\frac{q(c)k(c)}{k(c)-1}\right)^{\frac{rk(c)}{\gamma}}$$

B Predictable Hostility

We now analyze a setting where hostility arrives at some publicly known time T. This setup represents cases where countries can anticipate the arrival of hostility. Predictable hostility is the limiting case of anticipated geopolitical risk in which the post-T hostility hazard becomes arbitrarily large: as $\lambda \to \infty$, hostility begins at T with certainty. Thus, $T_h = T$.

From Lemma 2, the strategic firm's peace-time behavior matters for the government only through the induced distribution H_X^*. Thus, for an exposure path x^p with cumulative exposure X, the government's expected payoff is

$$W(x^p, H_X^*) = \int_0^{T_h} e^{-rt} x_t^p \, dt + e^{-rT_h} \Psi(X). \tag{16}$$

Compared to the anticipated-risk problem in equation (13), $V(X)$ is replaced by $\Psi(X)$, because hostility begins immediately at $T = T_h$. By Lemma 4, the government, therefore, implements any given $X \in [\phi T_h, T_h]$ by offering full exposure until $\tau(X)$ and minimal exposure thereafter. Substituting this path into equation (16), the government chooses cumulative exposure X to maximize

$$W(X, T_h) = \frac{1 - e^{-r\tau(X)}}{r} + \phi \frac{e^{-r\tau(X)} - e^{-rT_h}}{r} + e^{-rT_h} \Psi(X). \tag{17}$$

The following proposition characterizes the equilibrium exposure:

Proposition B.7. *Suppose hostility arrives at T_h that is known. Define $X_{T_h}(\tau) = \phi T_h + (1 - \phi)\tau$. Under Assumption 1, the government's equilibrium exposure path during peace is*

$$x_t^{p*} = \begin{cases} 1, & t < \tau^*, \\ \phi, & \tau^* \le t < T_h, \end{cases}$$

where,

$$\tau^* = \begin{cases} 0, & T_h < \underline{T}', \\ \tau^I(T_h), & \underline{T}' \le T_h < \overline{T}', \\ T_h, & T_h \ge \overline{T}'. \end{cases}$$

- *When* $\tau^I(T_h) \in (0, T_h)$ *exists, it is unique and satisfies*

$$\frac{(1-q_0)K}{r+\gamma} e^{-\mu X_{T_h}(\tau^I(T_h))}(1-\mu X_{T_h}(\tau^I(T_h))) = e^{r[T_h-\tau^I(T_h)]},$$

where $X^I(T_h) = X_{T_h}(\tau^I(T_h))$.

- *If* $\tau^I(T_h)$ *exists, the thresholds are* $\underline{T}' = \min\{T'_{IL}, T'_{FL}\}$ *and* $\overline{T}' = \max\{T'_{FL}, T'_{FI}\}$ *where* T'_{IL}, T'_{FL}, *and* T'_{FI} *uniquely solve,*

$$W(\phi T_h, T_h) = W(X_{T_h}(\tau^I(T_h)), T_h); W(T_h, T_h) = W(\phi T_h, T_h); W(T_h, T_h) = W(X_{T_h}(\tau^I(T_h)), T_h),$$

respectively. If $\tau^I(T_h)$ *does not exist, the thresholds are* $\underline{T}' = \overline{T}' = T'_{FL}$.

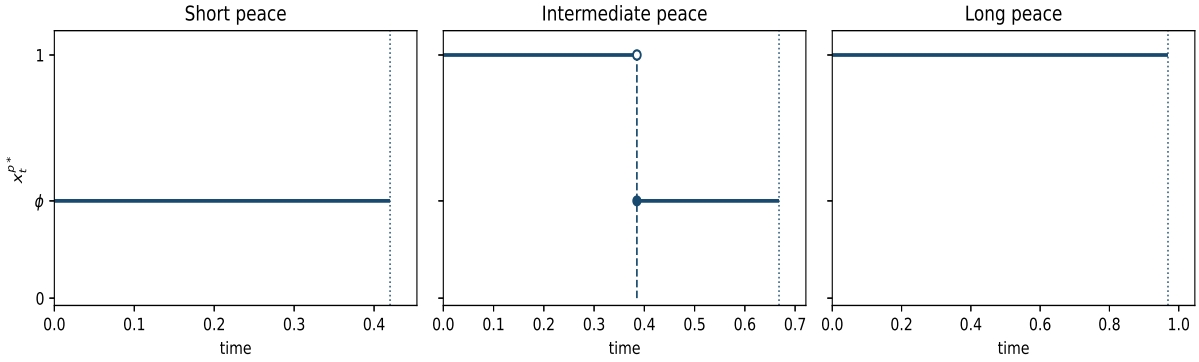


Figure 7: Peace-time stakes curve x_t^{p*} under different under short, intermediate, and long periods of predictable peace. Parameters: $r = 0.10$, $\gamma = 0.01$, $\phi = 0.40$, $q_0 = 0.90$, $K = 1.19$, and $\mu = 0.05$. The three panels correspond to $T_h = 0.420$ (short peace), $T_h = 0.668$ (intermediate peace), and $T_h = 0.969$ (long peace).

The structure of Proposition B.7 matches closely with that of Proposition 4. In fact, it is the limiting case of the anticipated geopolitical risk. First, note that as $\lambda \to \infty$, the second cutoff τ_2^* disappears. That is because $\tau_2^* \geq T$. In the predictable hostility case, hostility arrives almost surely at $T = T_h$. Thus, hostility arrives before the government could potentially liberalize during peace.

The first cutoff remains but its condition simplifies. In Proposition 4, the interior cutoff

satisfies

$$e^{-\frac{r+\lambda}{\phi}\left[\widetilde{X}-X_T(\tau^I(T))\right]}+\frac{\lambda}{\phi}\int_{X_T(\tau^I(T))}^{\widetilde{X}} e^{-\frac{r+\lambda}{\phi}\left[s-X_T(\tau^I(T))\right]}G(s)\,ds=e^{r\left(T-\tau^I(T)\right)}.$$

As $\lambda\to\infty$, the first term of the LHS vanishes and the second term converges to

$$\lim_{\lambda\to\infty}\left[\frac{\lambda}{\phi}\int_{X_T(\tau^I(T))}^{\widetilde{X}} e^{-\frac{r+\lambda}{\phi}\left(s-X_T(\tau^I(T))\right)}G(s)\,ds\right]=G(X_{T_h}(\tau^I(T_h))).$$

The second equality follows because the expression for G follows the first branch in equation (10) which yields the condition in Proposition B.7. The optimal predictable-hostility exposure path therefore restricts exposure throughout when T_h is short, grants full exposure before retrenching when T_h is intermediate, and maintains full exposure throughout when T_h is sufficiently long, as illustrated in Figure 7.

C General Baseline Harm

In this section, we consider a general harm specification. For a total capacity accumulated C, let the effective harm per unit of exposure x_t^h and undermining intensity β_t be

$$k(C) = a + KC, \tag{18}$$

where a denotes the baseline harm that does not depend on the accumulated capacity. The specification in the main text corresponds to $a = 1$, which is equal to the normalized marginal economic gain from exposure.

The government's flow payoff during hostility, conditional on the firm being adversarial with accumulated capacity C_{T_h} and remaining undetected until hostility T_h, is

$$\pi_t^h(x_t^h, \beta_t; C_{T_h}) = x_t^h - (a + KC_{T_h})\beta_t x_t^h.$$

We maintain the setup that the adversarial firm and host government play a zero-sum game during hostility, so the adversarial firm receives $-\pi_t^h(x_t^h, \beta_t; C_{T_h})$. The payoffs for the commercial firm and the government are aligned, and accordingly, we set $\beta_t = 0$ for the commercial firm.

This extension examines how the optimal exposure depends on the relative magnitude of baseline harm and economic gains. When $a \geq 1$, an adversarial firm can impose losses at least as large as the economic benefits from the exposure even without accumulating any capacity. Full undermining therefore guarantees the adversarial firm a non-negative hostility payoff. When $a < 1$, the government can benefit from retaining an adversarial firm with sufficiently low capacity even when the firm undermines with full intensity. We first analyze $a \geq 1$, before discussing the additional considerations that arise when $a < 1$.

High Baseline Harm ($a \geq 1$).

As we already cover $a = 1$ in our main analysis, we focus on the case when $a > 1$. For $a > 1$, full undermining yields a positive flow payoff to the adversarial firm even when it does not accumulate any capacity. Throughout this analysis, we maintain the following assumption:

Assumption 2. *The baseline effective harm satisfies*

$$a < 1 + \frac{K}{\mu}. \tag{19}$$

This assumption ensures that, in equilibrium, the adversarial firm always builds some positive capacity. To understand this, suppose $a \geq 1 + K/\mu$, then in the case when the government offers full exposure during hostility, the adversarial firm's capacity choice $C_Y^a = \frac{1}{\mu} - \frac{a-1}{K}$ becomes non-positive. Thus, the adversarial firm will never build any capacity given the access provided to her. When the government offers a testing contract during hostility, this assumption simplifies analysis by ensuring that the lower bound of the adversarial firm's equilibrium support is positive whenever cumulative exposure is positive.

Even when we generalize the harm, the host government's posterior belief that the firm is commercial given no peace-time detection, $q(c)$ is unchanged. However, full exposure during hostility now requires

$$q(c) \geq [1 - q(c)](a - 1 + Kc) \quad \Longleftrightarrow \quad (a - 1 + Kc)e^{-\mu c} \leq \frac{q_0}{1 - q_0}. \tag{20}$$

Thus, the function that captures the two effects of capacity on the government's hostility contract is

$$Y_a(c) = (a - 1 + Kc)e^{-\mu c}.$$

Importantly, $Y_a(0) = a - 1 > 0$ instead of $Y(0) = 0$. Consequently, unlike the case for $a = 1$, a testing contract can be optimal even when the government anticipates $c = 0$ because the baseline loss is greater than the marginal economic gain from exposure. This happens when

the prior belief is sufficiently low, i.e., $q_0 < \frac{a-1}{a}$. Therefore, this extension introduces a new region where the government offers a testing contract even when it believes the accumulated capacity is low.

When $q_0 \geq \frac{a-1}{a}$, the generalized harm extension essentially introduces an affine transformation for capacity. Define

$$h_a = \frac{a-1}{K}, \qquad d_a = q_0 + (1-q_0)e^{\mu h_a}, \qquad p_a = \frac{q_0}{d_a}.$$

We can transform the capacity $z = c + h_a$ so that $a + Kc = 1 + Kz$. Accordingly, $q_0 = d_a p_a$ and $(1-q_0)e^{-\mu c} = d_a(1-p_a)e^{-\mu z}$. The generalized hostility payoffs and marginal security costs are

$$\Psi_a(X) = d_a \Psi_1(X + h_a; p_a), \qquad G_a(X) = -\Psi_a'(X),$$

respectively, and the generalized cutoffs are $C_Y^a = 1/\mu - h_a$ and $C_H^a = C_H(p_a) - h_a$. Let $\widetilde{X}_a$ denote the solution to

$$-V_a'(X) = \lambda \int_0^\infty e^{-(r+\lambda)t} G_a(X+t)\, dt = 1.$$

The corresponding assumption for small ϕ, (Assumption 1), uses G_a and $\widetilde{X}_a$. Then, as the following proposition states formally, the optimal exposure paths retain the structure established in our main analysis with the capacity and thresholds adjusted for a.

Proposition C.8 (High Baseline Harm Equilibrium)**.** *Suppose Assumption 2 and the generalized version of Assumption 1 hold; and $q_0 \geq \frac{a-1}{a}$. Then, the optimal peace-time exposure paths offered by the government under persistent and anticipated geopolitical risk retain the characterizations in Proposition 3 and Proposition 4 respectively, where the characterizations use the generalized expressions for a given baseline harm a.*

The affine reduction also applies when $q_0 < (a-1)/a$. In this case, however, the prior that the firm is commercial is sufficiently low so that the government offers a testing contract even when it anticipates zero capacity building. Under persistent risk, the optimal exposure

path retains its earlier probationary form.

However, under anticipated geopolitical risk, this low prior rules out the initial high access that precedes restrictions in the U-shaped policy. Technically, when ϕ is sufficiently small, the low prior rules out an interior optimal cumulative exposure, i.e., $X_T^* \notin (\phi T, T)$. Consequently, the government either restricts access from the outset, liberalizing eventually when the cumulative exposure reaches $\widetilde{X}_a$, or grants full exposure throughout.

Low Baseline Harm $(a < 1)$.

When the baseline harm is low, i.e., $a < 1$, the government could benefit from retaining even an adversarial firm during hostility. Specifically, when the capacity accumulated by the adversarial firm until hostility is $C_{T_h} < \frac{1-a}{K}$, the government's flow payoff remains positive even when the firm undermines with full intensity.

However, because an adversarial firm and the host government play a zero-sum game during hostility, the adversarial firm's flow payoffs are negative when $C_{T_h} < \frac{1-a}{K}$ even when $\beta_t = 1$. Since the firm's outside option is 0, the firm would prefer to exit the host country's market. This introduces an additional consideration when designing the optimal exposure: restricting exposure may limit capacity accumulation sufficiently to induce an adversarial firm to exit when hostility begins. A full analysis of the optimal exposure in this case requires modifications to the model to allow for voluntary firm exit. We leave that to future work.

D Proofs[26]

Proof of Lemma 1. Suppose the government offers a peace-time contract, $\mathcal{C}^p = (x^p, T_R^p)$, that specifies exclusion at some finite T_R^p while peace continues. Consider an alternate contract that offers the same access path but never excludes the firm during peace, $\mathcal{C}^{p'} = (x^p, T_R^{p'})$ where $T_R^{p'} = \infty$. For any $T_h > T_R^p$, the government can then exclude the firm at the start of hostility. Then, because capacity produces no harm during peace, $\mathcal{C}^p$ provides the government no additional security benefits compared to $\mathcal{C}^{p'}$ irrespective of the firm's efforts. Additionally, peace-time economic gains are also weakly lower under $\mathcal{C}^p$ compared to $\mathcal{C}^{p'}$. Thus, the government can do strictly better by offering $\mathcal{C}^{p'}$ and excluding at the start of hostility.

For $T_h \leq T_R^p$, the government can still offer $\mathcal{C}^{p'}$ and, at T_h offer the same hostility contract it would have offered following $\mathcal{C}^p$. The government cannot do worse under this new contract. Because T_h is random, there exists a strictly profitable deviation for the government in which it offers a peace-time contract that does not exclude during peace. Therefore, every finite peace-time exclusion is strictly sub-optimal and $T_R^{p*} = \infty$.

□

Proof of Proposition 1. Conditional on no peace-time detection, when the government believes that the capacity stock c, its posterior is

$$q(c) = \frac{q_0}{q_0 + (1 - q_0)e^{-\mu c}}.$$

Given $k(c) = 1 + Kc$, from Proposition A.6, the government grants immediate full exposure if and only if

$$q(c) \geq \frac{k(c) - 1}{k(c)} = \frac{Kc}{1 + Kc},$$

which is equivalent to

$$\begin{aligned} q_0(1 + Kc) &\geq Kc\left[q_0 + (1 - q_0)e^{-\mu c}\right] \\ \frac{q_0}{1 - q_0} &\geq Kce^{-\mu c}. \end{aligned}$$

Recall that $Y(c) = Kce^{-\mu c}$. Its derivative is $Y'(c) = Ke^{-\mu c}(1 - \mu c)$. Hence, Y is strictly

[26] I am currently revising the proofs and would appreciate any comments.

increasing on $[0, 1/\mu)$ and strictly decreasing on $(1/\mu, \infty)$. Thus, its unique maximizer is $C_Y = 1/\mu$ with $Y(C_Y) = \frac{K}{\mu e}$.

Consider two cases:

- If $\frac{K}{\mu e} \leq \frac{q_0}{1-q_0}$, then $Y(c) \leq q_0/(1-q_0)$ for every $c \geq 0$, so the government grants immediate full exposure for every c.
- If $\frac{K}{\mu e} \geq \frac{q_0}{1-q_0}$, because $Y(0) = 0 < \frac{q_0}{1-q_0} \leq Y(C_Y)$, continuity and strict monotonicity imply that there is a unique $c^- \in (0, C_Y]$ satisfying $Y(c^-) = \frac{q_0}{1-q_0}$. Similarly, since Y is strictly decreasing above C_Y and converges to zero, there is a unique $c^+ \in [C_Y, \infty)$ satisfying $Y(c^+) = \frac{q_0}{1-q_0}$. Thus,

$$Y(c) \leq \frac{q_0}{1-q_0} \iff c \leq c^- \text{ or } c \geq c^+.$$

  When $\frac{K}{\mu e} = \frac{q_0}{1-q_0}$, $Y(C_Y) = \frac{q_0}{1-q_0}$ and single-peakedness of $Y(c)$ implies that $Y(c) < Y(C_Y) = \frac{q_0}{1-q_0}$ for all $c \neq C_Y$. Thus, $c^- = C_Y = c^+$.

  Therefore, the government therefore grants immediate full exposure for $c \leq c^-$ and $c \geq c^+$, and offers the screening contract for $c \in [c^-, c^+]$.

If capacity building is detected during peace, the government learns that the firm is adversarial and therefore excludes it when hostility begins.

□

Proof of Proposition 2. We prove this in two parts. First consider the following Lemma:

Lemma D.5 (Equilibrium capacity distribution)**.** *Suppose the government provides cumulative peace-time exposure X. The equilibrium capacity distribution is*

$$H_X^*(c) = \begin{cases} 0, & c < \min\{X, C_Y\}, \\ \dfrac{1 + K\big(c - \min\{X, C_Y\}\big)}{M\big(\min\{X, C_H\}\big)}, & \min\{X, C_Y\} \leq c < \min\{X, C_H\}, \\ 1, & c \geq \min\{X, C_H\}, \end{cases}$$

where $M(z) = (1 + Kz)\big(1 - q(z)\big)$ and $C_H = \min\{c \geq C_Y : \ \mu q(c)(c + \frac{1}{K}) \geq 1\}$.

Proof. Fix a total exposure budget X. Define

$$w(c) = \mu q(c)\left(c + \frac{1}{K}\right) \tag{21}$$

Because $q(c)$ is strictly increasing, $w(c)$ is strictly increasing. Moreover,

$$w(C_Y) \geq 1 \quad \Longleftrightarrow \quad \frac{K}{\mu e} \leq \frac{q_0}{1-q_0}.$$

Define

$$C_H = \min\{c \geq C_Y : w(c) \geq 1\}. \tag{22}$$

We have $C_H = C_Y$ in the no-screening case. In the strict screening case, $C_H > C_Y$ and is the unique solution to $w(C_H) = 1$. Then consider the following cases.

First, suppose $Y(C_Y) \leq q_0/(1-q_0)$. For any $C \leq X$, the strategic type's continuation payoff is proportional to

$$e^{-\mu C}\big(k(C) - 1\big) = Y(C).$$

Because Y is strictly increasing on $[0, C_Y]$ and strictly decreasing on (C_Y, ∞), the unique best response on $[0, X]$ is $C^* = \min\{X, C_Y\}$. Thus, the unique induced distribution is $H_X^* = \delta_{\min\{X, C_Y\}}$.

Next, suppose $Y(C_Y) > q_0/(1-q_0)$. First suppose $X \leq C_Y$. For any $C \leq X$, the strategic type's continuation payoff is proportional to

$$e^{-\mu C}\big(k(C) - 1\big) = Y(C).$$

Because Y is strictly increasing on $[0, C_Y]$, the unique best response on $[0, X]$ is $C^* = X$. Thus, the unique induced distribution is $H_X^* = \delta_X$.

Next, suppose $X \geq C_Y$. This part is similar to the problem we study in Zalke (2026). The calculations for the optimal hostility contract as well as the distribution over C follow the same way as in the screening problem studied in Zalke (2026). Thus, the proof here shows only the calculations that are relevant to the proof here.

The best response cannot choose a total capacity less than C_Y because her continuation payoff, $e^{-\mu C}\big(k(C) - 1\big) = Y(C)$, is increasing until C_Y. So, the support of strategic type's best response must have a lower bound at C_Y.

Consider a distribution with support $[C_Y, z]$ where $z \leq X$. Let the distribution be of the form

$$dH(C) = m_Y \delta_{C_Y}(dC) + h(C)\mathbf{1}\{C \in (C_Y, z)\}\, dC + m_z \delta_z(dC),$$

where $h(C)$ denotes the density on (C_Y, z). For $C \in (C_Y, z)$, define the posterior survivor mass as the probability, conditional on no detection, that the firm is either commercial or a strategic type with total capacity strictly greater than C as

$$s_H(C) = \frac{q_0 + (1 - q_0) \int_C^z e^{-\mu y}\, dH(y)}{\Delta_H}, \tag{23}$$

where $\Delta_H = Pr(\text{no detect}|H) = q_0 + (1 - q_0) \int e^{-\mu C}\, dH(C)$.

The strategic firm must get the same expected payoff over the support in its best response. Thus, for $C \in (C_Y, z)$, it must be that $e^{-\mu C} u^h(\mathcal{C}^h(H), C)$ is constant. The fixed-posterior screening equations derived in Zalke (2026) imply that for all $C \in (C_Y, z)$, this indifference condition can be written in terms of the posterior survivor mass as

$$s_H(C) = s_H(C_Y^+) e^{-\mu(C - C_Y)}. \tag{24}$$

At the lower endpoint, the government's optimal hostility contract must satisfy $k(C_Y)(1 - s_H(C_Y^+)) = 1$. This is explained in detail in the proof of Theorem 2 in Zalke (2026). Thus,

$$s_H(C_Y^+) = \frac{k(C_Y) - 1}{k(C_Y)} = \frac{K/\mu}{1 + K/\mu} = \frac{K}{K + \mu}. \tag{25}$$

The second equality follows because $k(C) = 1 + KC$ and $C_Y = 1/\mu$. Combining this with (23), the atom at the lower bound, C_Y, of the support of the posterior distribution is

$$1 - s_H(C_Y^+) = 1 - \frac{q_0 + (1 - q_0) \int_{C_Y^+}^z e^{-\mu y}\, dH(y)}{\Delta_H} = \frac{(1 - q_0) m_Y e^{-\mu C_Y}}{\Delta_H} = \frac{\mu}{K + \mu}. \tag{26}$$

Next, for $C \in (C_Y, z)$, from (23),

$$s'_H(C) = -\frac{(1 - q_0) h(C) e^{-\mu C}}{\Delta_H} = -\mu s_H(C),$$

where the second equality follows from differentiating equation (24). Using (24) and $s_H(C_Y^+) =$

$K/(K+\mu)$,

$$h(C) = \frac{\mu\Delta_H}{1-q_0}e^{\mu C}s_H(C) = \frac{\mu\Delta_H}{1-q_0}e^{\mu C}\frac{K}{K+\mu}e^{-\mu(C-C_Y)}$$
$$= \frac{\mu K}{K+\mu}\frac{\Delta_H e^{\mu C_Y}}{1-q_0} = Km_Y. \tag{27}$$

The last equality follows because equation (26) implies

$$m_Y = \frac{\mu}{\mu+K}\frac{\Delta_H e^{\mu C_Y}}{1-q_0}.$$

Therefore, for $C \in (C_Y, z)$,

$$h(C) = Km_Y. \tag{28}$$

For the atom at the upper bound of the support, z, suppose $m_z = m_Y l$. Because the total probability must be equal to one,

$$m_Y + Km_Y(z - C_Y) + m_Y l = 1,$$

which implies that

$$m_Y = \frac{1}{1 + K(z - C_Y) + l}. \tag{29}$$

From (23), the posterior survivor mass just below z is

$$s_H(z^-) = \frac{q_0 + (1-q_0)m_z e^{-\mu z}}{\Delta_H} = \frac{K}{K+\mu}e^{-\mu(z-C_Y)},$$

where the second equality follows from equation (24). It follows from equation (26) that

$$\Delta_H = (1-q_0)m_Y e^{-\mu C_Y}\frac{K+\mu}{\mu}.$$

Equating the two expressions for $s_H(z^-)$ and substituting Δ_H,

$$\begin{aligned}
q_0 + (1-q_0)m_Y l e^{-\mu z} &= (1-q_0)m_Y \frac{K}{\mu} e^{-\mu z} \\
q_0 &= (1-q_0)m_Y e^{-\mu z}\left(\frac{K}{\mu} - l\right) \\
q_0 &= (1-q_0)\frac{1}{1+K(z-C_Y)+l} e^{-\mu z}\left(\frac{K}{\mu} - l\right) \qquad \text{(from equation (29))} \\
l\big(q_0 + (1-q_0)e^{-\mu z}\big) &= (1-q_0)\frac{K}{\mu} e^{-\mu z} - q_0\big(1+K(z-C_Y)\big).
\end{aligned}$$

Therefore,

$$\begin{aligned}
l(z) &= \frac{(1-q_0)\frac{K}{\mu}e^{-\mu z} - q_0\big(1+K(z-C_Y)\big)}{q_0 + (1-q_0)e^{-\mu z}} \\
&= \frac{K}{\mu}\left[1 - \mu q(z)\left(z + \frac{1}{K}\right)\right]. \qquad (30)
\end{aligned}$$

Consider the numerator in the first line of (30). Denote it by $N(z)$. At $z = C_Y$,

$$N(C_Y) = (1-q_0)\frac{K}{\mu}e^{-\mu C_Y} - q_0.$$

Because $Y(C_Y) = e^{-\mu C_Y}K/\mu$ and $Y(C_Y) > q_0/(1-q_0)$, $N(C_Y) > 0$. Moreover,

$$N'(z) = -(1-q_0)Ke^{-\mu z} - q_0 K < 0.$$

So N is strictly decreasing. Thus, there exists a unique $C_H > C_Y$ such that $l(C_H) = 0$. From (30), this is equivalent to

$$q(C_H)\left(C_H + \frac{1}{K}\right) = \frac{1}{\mu}.$$

When $X \geq C_H$, the optimal hostility contract offered by the government and the strategic firm's choice over C are given by the expressions in Theorem 2 in Zalke (2026). Thus, the strategic firm's equilibrium distribution is of the form

$$dH_X^*(C) = \frac{1}{1+K(C_H - C_Y)}\delta_{C_Y}(dC) + \frac{K}{1+K(C_H-C_Y)}\mathbf{1}\{C \in (C_Y, C_H]\}\, dC. \qquad (31)$$

We can denote the stakes in terms of what capacity is screened at time t. Let κ_t denote

the total capacity that is indifferent (or screened) at time t. That is, the agent who built capacity κ_t is indifferent between undermining and not at time t. Thus, the optimal stakes curve during hostility follows:

- The cutoff starts at screening capacity C_Y, and the cutoff path satisfies
$$\kappa_t = \frac{1}{K}\left(\left[(1+KC_Y)^2 + \frac{2\gamma K}{\mu}t\right]^{1/2} - 1\right).$$

- The stakes curve is
$$x_t^h = \begin{cases} \dfrac{\gamma K \kappa_t}{(r+\gamma)(1+K\kappa_t)} e^{-\mu(C_H-\kappa_t)\left[1+\frac{r}{2\gamma}(2+K(C_H+\kappa_t))\right]}, & t < \dfrac{\mu}{2\gamma K}\left[(1+KC_H)^2-(1+KC_Y)^2\right], \\ 1, & t \geq \dfrac{\mu}{2\gamma K}\left[(1+KC_H)^2-(1+KC_Y)^2\right]. \end{cases}$$

When $X < C_H$, the indifference equations remain the same along the C_H envelope at the end before it jumps. In this case, $z = X$ and denote

$$M(X) = 1 + K(X - C_Y) + l(X) = (1+KX)(1-q(X)).$$

From the calculations above, the distribution over total capacity must be of the form

$$dH_X^*(C) = \frac{1}{M(X)}\delta_{C_Y}(dC) + \frac{K}{M(X)}\mathbf{1}\{C \in (C_Y, X)\}\, dC + \frac{l(X)}{M(X)}\delta_X(dC). \tag{32}$$

The hostility contract associated with this H_X^* is the truncated version of the hidden-harm moving-cutoff contract. That is, the stakes are similar as above so that screening starts at C_Y and evolves according to the same expression so that

$$\kappa_t = \frac{1}{K}\left(\left[(1+KC_Y)^2 + \frac{2\gamma K}{\mu}t\right]^{1/2} - 1\right), \tag{33}$$

with the cutoff reaching $z = X$ at

$$T(X) = \frac{\mu}{2\gamma K}\left[(1+KX)^2 - (1+KC_Y)^2\right]. \tag{34}$$

However, because of the atom at $z = X$, the optimal stakes continue along the X-indifference envelope. This follows from the results in Kolb and Madsen (2023). The stakes continue

along this path until it jumps to one and the duration of this follows from the indifference of the continuation value just before the stakes jump to 1. Thus, the duration of this is

$$T_A(X) = \frac{1+KX}{\gamma} \log\left(\frac{q_0 + (1-q_0)e^{-\mu X}}{q_0 \mu \left(X + \frac{1}{K}\right)}\right). \tag{35}$$

The associated hostility stakes curve is therefore

$$x_t^h(X) = \begin{cases} \frac{\gamma K \kappa_t}{(r+\gamma)(1+K\kappa_t)} e^{\left\{-\mu(X-\kappa_t) - \frac{r\mu}{2\gamma K}\left[(1+KX)^2 - (1+K\kappa_t)^2\right] - \left[r + \frac{\gamma}{1+KX}\right] T_A(X)\right\}}, & 0 \leq t < T(X), \\ \frac{\gamma K X}{(r+\gamma)(1+KX)} e^{-\left(r + \frac{\gamma}{1+KX}\right)(T_A(X) - (t - T(X)))}, & t \in [T(X), T(X) + T_A(X)) \\ 1, & t \geq T(X) + T_A(X). \end{cases} \tag{36}$$

Note that the strategic firm is indifferent over this support and she has no profitable deviation outside of the support $[C_Y, \min\{X, C_H\}]$. The government's hostility contract is the optimal contract it can offer given the distribution over capacity. Thus, this is an equilibrium.

The proof for uniqueness is the same as the proof for uniqueness in Theorem 2 in Zalke (2026). First note that we showed above that the lower bound of the support of the distribution over C must be above C_Y whenever $X \geq C_Y$. Thus, it follows from the calculations above that the strategic type will never choose a capacity below C_Y. When $C_Y < X < C_H$, feasibility rules out choosing a capacity greater than X. For $X > C_H$, the strategic firm can feasibly choose a $C > C_H$; however, any such deviation gives the strategic firm a strictly lower payoff.

Finally, using the same approach as in the proof of Theorem 2 in Zalke (2026), there cannot be any equilibria where the distribution over C has an interior atom or a gap in the support. Suppose, by contradiction, there exists such an equilibrium with an interior atom. Suppose H has an atom of size $m > 0$ at some $C_0 \in (C_Y, z)$. In such an equilibrium, the hostility contract that the government offers must wait on the C_0 indifference envelope. However, in that case, the strategic type who builds a capacity of C_0 can do better by instead building a capacity of $C_0 + \epsilon$. Thus, this cannot be an equilibrium.

To rule out a gap in the support, suppose there is a gap (a, b) in the equilibrium distribution. It follows that the strategic type must get a strictly lower payoff for choosing a capacity $C \in (a, b)$. Indifference implies that $e^{-\mu a} U^N(a) = e^{-\mu b} U^N(b) \implies U^N(b) = e^{\mu(b-a)} U^N(a)$. Also, $U^N(C)$ is affine over the gap. However, the RHS is strictly convex. So there must exist

some $C \in (a,b)$ such that $e^{-\mu C}U^N(C) > e^{-\mu a}U^N(a)$, which implies that the strategic firm has a profitable deviation. Thus, this cannot be an equilibrium.

Stating the distribution as a CDF, we get

$$H_X^*(c) = \begin{cases} 0, & c < \min\{X, C_Y\}, \\ \dfrac{1 + K\big(c - \min\{X, C_Y\}\big)}{M\big(\min\{X, C_H\}\big)}, & \min\{X, C_Y\} \le c < \min\{X, C_H\}, \\ 1, & c \ge \min\{X, C_H\}, \end{cases}$$

where $M(z) = (1 + Kz)\big(1 - q(z)\big)$ and $C_H = \min\{c \ge C_Y : \ \mu q(c)(c + \frac{1}{K}) \ge 1\}$.

This completes the proof.

□

Next, consider the following lemma:

Lemma D.6. *Write $\kappa = \min\{X, C_Y\}$ and $z = \min\{X, C_H\}$, and let*

$$D(X) = \frac{\mu}{2\gamma K}\Big[(1 + Kz)^2 - (1 + K\kappa)^2\Big] - \frac{1 + Kz}{\gamma}\log\frac{w(z)}{\mu\kappa}, \qquad w(c) = \mu q(c)\Big(c + \frac{1}{K}\Big).$$

The government's expected hostility payoff as a function of total exposure X is

$$\Psi(X) = \begin{cases} \dfrac{q_0}{r} - \dfrac{1 - q_0}{r + \gamma} K\kappa e^{-\mu\kappa}, & X \le c^-, \\ \dfrac{q_0\gamma}{r(r + \gamma)} e^{-rD(X)}, & X \ge c^-, \end{cases}$$

where c^- is as characterized in Proposition 1, with $c^- = \infty$ when $K/\mu e \le q_0/(1 - q_0)$.

Proof. Fix a total exposure X. Define $w(c)$ and C_H as we do in equations (21) and (22), respectively.

First, suppose $\frac{K}{\mu e} \le \frac{q_0}{1 - q_0}$. By Lemma D.5, the adversarial firm's equilibrium capacity distribution is degenerate at $\kappa = \min\Big\{X, \frac{1}{\mu}\Big\}$. Moreover, from Proposition 1, the government

grants full exposure during hostility for every inherited capacity. Hence,

$$\begin{aligned}\Psi(X) &= \left(q_0 + (1-q_0)e^{-\mu\kappa}\right) V^h\left(\mathcal{C}^{h*}(q(\kappa), k(\kappa)), \kappa\right) \\ &= \left(q_0 + (1-q_0)e^{-\mu\kappa}\right)\left[\frac{q(\kappa)}{r} - (1-q(\kappa))\frac{k(\kappa)-1}{r+\gamma}\right] \\ &= \frac{q_0}{r} - \frac{1-q_0}{r+\gamma}\kappa K^{-\mu\kappa} = \frac{q_0}{r} - \frac{1-q_0}{r+\gamma} K \min\left\{X, \frac{1}{\mu}\right\} e^{-\mu\min\{X, 1/\mu\}}.\end{aligned}$$

Next, suppose $\frac{K}{\mu e} > \frac{q_0}{1-q_0}$. First, consider $X \leq C_Y = 1/\mu$. By Lemma D.5, the adversarial firm's equilibrium capacity distribution is degenerate at X. Hence, the government enters hostility with posterior

$$q(X) = \frac{q_0}{q_0 + (1-q_0)e^{-\mu X}}$$

and known effective harm $k(X) = 1 + KX$.

For $X \leq c^-$, the government grants full exposure during hostility. Therefore,

$$\begin{aligned}\Psi(X) &= \left(q_0 + (1-q_0)e^{-\mu X}\right) V^h\left(\mathcal{C}^{h*}(q(X), k(X)), X\right) \\ &= \left(q_0 + (1-q_0)e^{-\mu X}\right)\left[\frac{q(X)}{r} - (1-q(X))\frac{k(X)-1}{r+\gamma}\right] \\ &= \frac{q_0}{r} - \frac{(1-q_0)e^{-\mu X}KX}{r+\gamma}.\end{aligned}$$

Now suppose $c^- < X \leq C_Y$. The induced capacity distribution is still degenerate at X, but the government now offers the low-posterior screening contract. Hence, from Proposition A.6,

$$\begin{aligned}\Psi(X) &= \left(q_0 + (1-q_0)e^{-\mu X}\right) V^h\left(\mathcal{C}^{h*}(q(X), k(X)), X\right) \\ &= \left(q_0 + (1-q_0)e^{-\mu X}\right) q(X)\frac{\gamma}{r(r+\gamma)}\left(\frac{q(X)k(X)}{k(X)-1}\right)^{rk(X)/\gamma} \\ &= \frac{q_0\gamma}{r(r+\gamma)}\left(\frac{q(X)(1+KX)}{KX}\right)^{r(1+KX)/\gamma} \\ &= \frac{q_0\gamma}{r(r+\gamma)}\left(\frac{q_0(1+KX)}{\left[q_0 + (1-q_0)e^{-\mu X}\right]KX}\right)^{r(1+KX)/\gamma}.\end{aligned}$$

Finally, consider $X > C_Y$ and set $z = \min\{X, C_H\}$. Lemma D.5 gives the adversarial firm's equilibrium distribution over capacity, and its proof characterizes the corresponding optimal

hostility contract. The moving cutoff reaches z after

$$\begin{aligned} T(z) &= \frac{\mu}{2\gamma K}\left[(1+Kz)^2-(1+KC_Y)^2\right] \\ &= \frac{\mu}{\gamma}\left[(z-C_Y)+\frac{K}{2}(z^2-C_Y^2)\right]. \end{aligned}$$

When $z < C_H$, the atom at z requires the hostility contract to remain on the z-indifference envelope before exposure jumps to one. Let q_z denote the posterior probability that the firm is commercial at the beginning of this interval. Then,

$$q_z = \frac{q_0}{q_0 + (1-q_0)\frac{l(z)}{M(z)}e^{-\mu z}}.$$

From the expressions for $l(z)$ and $M(z)$ in the proof of Lemma D.5,

$$\frac{l(z)}{M(z)} = \frac{(1-q_0)\frac{K}{\mu}e^{-\mu z} - q_0\left[1+K(z-C_Y)\right]}{(1-q_0)e^{-\mu z}(1+Kz)}.$$

Substituting this expression into the posterior gives

$$q_z = \frac{q_0\mu\left(z+\frac{1}{K}\right)}{q_0+(1-q_0)e^{-\mu z}}.$$

Therefore, the additional delay generated by the atom at z is

$$T_A(z) = \frac{1+Kz}{\gamma}\log\left(\frac{1}{q_z}\right) = \frac{1+Kz}{\gamma}\log\left(\frac{q_0+(1-q_0)e^{-\mu z}}{q_0\mu\left(z+\frac{1}{K}\right)}\right).$$

When $z = C_H$, $q(C_H)\left(C_H+\frac{1}{K}\right) = \frac{1}{\mu}$. Thus, $T_A(C_H) = 0$.

Thus, the total time before the hostility contract grants full exposure is $D(z) = T(z)+T_A(z)$. Substituting $C_Y = 1/\mu$ gives

$$D(z) = \frac{\mu}{\gamma}\left(z-\frac{1}{\mu}\right)\left[1+\frac{K}{2}\left(z+\frac{1}{\mu}\right)\right] + \frac{1+Kz}{\gamma}\log\left(\frac{q_0+(1-q_0)e^{-\mu z}}{q_0\mu\left(z+\frac{1}{K}\right)}\right).$$

As we compute in the proof of Theorem 2 in Zalke (2026), the government's hostility payoff

in equilibrium is

$$\Psi(X) = \frac{q_0\gamma}{r(r+\gamma)} e^{-rD(\min\{X, C_H\})}.$$

We can define $\underline{c}_X = \min\{X, C_Y\}$ and $\bar{c}_X = \min\{X, C_H\}$. Substituting these into $D(X)$ yields and let

$$D(X) = \frac{\mu}{2\gamma K}\left[(1+K\bar{c}_X)^2 - (1+K\underline{c}_X)^2\right] - \frac{1+K\bar{c}_X}{\gamma}\log\frac{w(\bar{c}_X)}{\mu\underline{c}_X}, \tag{37}$$

where $w(\cdot)$ is as defined in (21). Combining the three cases and simplifying using $\underline{c}_X$, $\bar{c}_X$ and (37),

$$\Psi(X) = \begin{cases} \dfrac{q_0}{r} - \dfrac{1-q_0}{r+\gamma} K\underline{c}_X e^{-\mu\underline{c}_X}, & X \le c^-, \\ \dfrac{q_0\gamma}{r(r+\gamma)} e^{-rD(X)}, & X \ge c^-, \end{cases}$$

where c^- is as characterised in Proposition 1, with $c^- = \infty$ when $K/\mu e \le q_0/(1-q_0)$.

This completes the proof.

□

The two lemmas prove both the parts of the proposition. □

Proof of Lemma 2. Fix a peace-time exposure path $x^p = (x^p_t)_{t\ge 0}$, and let $X_t = \int_0^t x^p_s\, ds$. We begin by showing existence of a feasible capacity process $(C^*_t)_{t\ge 0}$, proving the first part of the lemma. Consider the following cases:

Case 1: Suppose $\frac{K}{\mu e} \le \frac{q_0}{1-q_0}$.
From Proposition 2, $C_H = \min\{c \ge C_Y : w(c) \ge 1\}$. In this case,

$$w(C_Y) = \mu\frac{q_0}{q_0 + \frac{1-q_0}{e}}\left(\frac{1}{\mu} + \frac{1}{K}\right) = \frac{1}{1+\frac{1-q_0}{q_0 e}}\left(1+\frac{\mu}{K}\right) \ge \frac{1}{1+\frac{1-q_0}{q_0 e}}\left(1+\frac{1-q_0}{q_0 e}\right) = 1.$$

The inequality follows from $\frac{K}{\mu e} \le \frac{q_0}{1-q_0}$. Thus, $C_H = C_Y$. It follows from Proposition 2 that $\underline{c}_X = \bar{c}_X = \min\{X, C_Y\}$ and $H^*_X(c) = \mathbf{1}_{c\ge \min\{X, C_Y\}}$. Set $C^*_t = \min\{X_t, C_Y\}$. Then, $C^*_t \sim H^*_{X_t}$ for every t and $0 \le \dot{C}^*_t \le x^p_t$ a.e..

Case 2: Suppose $\frac{K}{\mu e} > \frac{q_0}{1-q_0}$.

For $X_t \leq C_Y$, set $C_t^* = X_t$. The firm can accumulate this capacity by choosing $\alpha_t = 1\ \forall t$ such that $X_t \leq C_Y$.

For $X_t > C_Y$, consider the following capacity process

$$C_t^*(u) = \min\left\{\overline{c}_{X_t}, \underline{c}_{X_t} + \frac{\max\{uM(\overline{c}_{X_t}) - 1, 0\}}{K}\right\}, \tag{38}$$

where the firm draws $u \sim U[0,1]$ at t when $X_t = C_Y$. Then, for $\underline{c}_{X_t} \leq c < \overline{c}_{X_t}$,

$$\begin{aligned}
\Pr(C_t^* \leq c) &= \Pr\left(\max\{uM(\overline{c}_{X_t}) - 1, 0\} \leq K(c - \underline{c}_{X_t})\right) \\
&= \Pr\left(uM(\overline{c}_{X_t}) - 1 \leq K(c - \underline{c}_{X_t})\right) \\
&= \Pr\left(u \leq \frac{1 + K(c - \underline{c}_{X_t})}{M(\overline{c}_{X_t})}\right) = \frac{1 + K(c - \underline{c}_{X_t})}{M(\overline{c}_{X_t})}.
\end{aligned}$$

From (38), it follows that $\Pr(C_t^* \leq \overline{c}_{X_t}) = 1$. Thus, $C_t^* \sim H_{X_t}^*$ for every t.

To see this capacity process is feasible, consider any realized path $C^*(u)$. For $\underline{c}_{X_t} \leq C_t^*(u) \leq \overline{c}_{X_t}$, a.e.,

$$\alpha_t(u) = \begin{cases} 0, & C_t^*(u) = \underline{c}_{X_t}, \\ \dfrac{uM'(\overline{c}_{X_t})}{K}, & \underline{c}_{X_t} < C_t^*(u) < \overline{c}_{X_t}, \\ \mathbf{1}_{\{X_t < C_H\}}, & C_t^*(u) = \overline{c}_{X_t}, \end{cases}$$

For $\underline{c}_{X_t} < C_t^*(u) < \overline{c}_{X_t}$,

$$\begin{aligned}
\alpha_t(u) = \frac{\dot{C}_t^*(u)}{x_t^p} = u\frac{M'(\overline{c}_{X_t})}{K} &= \frac{u}{K}\left(K(1 - q(\overline{c}_{X_t})) - q'(\overline{c}_{X_t})(1 + K\overline{c}_{X_t})\right) \\
&= u\left(1 - q(\overline{c}_{X_t}) - q'(\overline{c}_{X_t})\left(\frac{1}{K} + \overline{c}_{X_t}\right)\right) \\
&= u\left(1 - q(\overline{c}_{X_t}) - \mu q(\overline{c}_{X_t})(1 - q(\overline{c}_{X_t}))\left(\frac{1}{K} + \overline{c}_{X_t}\right)\right) \\
&= u\left(\left(1 - q(\overline{c}_{X_t})\right)\left(1 - w(\overline{c}_{X_t})\right)\right).
\end{aligned}$$

From the definition of C_H, $w(\overline{c}_{X_t}) \leq 1$. Thus, $\alpha_t(u) \in [0,1]$ and $\dot{C}_t^*(u) = \alpha_t(u)x_t^p$ a.e.. Since $C_0^*(u) = 0$, and $C^*(u)$ is absolutely continuous a.e., it follows that $C_t^*(u) = \int_0^t \alpha_s(u)x_s^p\, ds$ for every t. For the capacity path $(C_t^*)_{t\geq 0}$ is feasible.

Next, fix any PBE following $\mathcal{C}^p$. Let H_t denote the adversarial firm's capacity distribution

at t in that PBE and let $\mathcal{C}_t^h$ denote the government's hostility contract when hostility arrives at t. Then, at every t, the government's ex-ante hostility payoff is

$$U^G(H_t, \mathcal{C}_t^h; X_t) \geq U^G(H_t, \mathcal{C}_{X_t}^{h*}; X_t) \geq U^G(H_{X_t}^*, \mathcal{C}_{X_t}^{h*}; X_t) \geq U^G(H_{X_t}^*, \mathcal{C}_t^h; X_t), \tag{39}$$

where $H_{X_t}^*$ is the distribution in Proposition 2 and $\mathcal{C}_{X_t}^{h*}$ is the corresponding hostility contract in that PBE. First inequality follows because $\mathcal{C}_t^h$ is optimal given H_t, the second follows because the adversarial firm does better with $H_{X_t}^*$ against a hostility contract $\mathcal{C}_{X_t}^{h*}$ which implies the government does worse because of the zero-sum game. The third follows because $\mathcal{C}_{X_t}^{h*}$ is optimal given $H_{X_t}^*$.

Thus, for the adversarial firm,

$$U^F(H_{X_t}^*, \mathcal{C}_t^h; X_t) \geq U^F(H_t, \mathcal{C}_t^h; X_t), \tag{40}$$

with the inequality being strict if $H_t \neq H_{X_t}^*$ for any t. If (40) is an equality, all the weak inequalities in (39) are equalities and $H_t = H_{X_t}^*$.

Next, consider the case if $H_t \neq H_{X_t}^*$ for some t. Because both H_t and $H_{X_t}^*$ follow from capacity process that are absolutely continuous ($\because C_t = \int_0^t \alpha_s x_s^p \, ds$), the distributions that they induce are weakly continuous in t. Therefore, if there exists any t' such that $H_{t'} \neq H_{X_{t'}}^*$, there exists a positive measure of t around t' such that $H_t \neq H_{X_t}^*$. It follows from (40) that the adversarial firm has a profitable deviation to $C_t^* \sim H_{X_t}^*$. Thus, this cannot be a PBE. Therefore, if $H_t \neq H_{X_t}^*$ for any positive measure of Therefore, in any PBE, $C_t^* \sim H_{X_t}^*$ at every t.

□

Proof of Lemma 3. From Proposition 2, the government's ex ante hostility payoff is

$$\Psi(X) = \begin{cases} \dfrac{q_0}{r} - \dfrac{1-q_0}{r+\gamma} K \underline{c}_X e^{-\mu \underline{c}_X}, & X \leq c^-, \\ \dfrac{q_0 \gamma}{r(r+\gamma)} e^{-rD(X)}, & X \geq c^-, \end{cases} \tag{41}$$

where c^- is as characterized in Proposition 1, with $c^- = \infty$ when $K/\mu e \leq q_0/(1-q_0)$, and

$$D(X) = \frac{\mu}{2\gamma K}\left[(1+K\bar{c}_X)^2 - (1+K\underline{c}_X)^2\right] - \frac{1+K\bar{c}_X}{\gamma} \log \frac{w(\bar{c}_X)}{\mu \underline{c}_X}.$$

Consider the following cases:

Case 1: When $X \leq c^-$
From (41),

$$\Psi(X) = \frac{q_0}{r} - \frac{1-q_0}{r+\gamma} K \underline{c}_X e^{-\mu \underline{c}_X},$$

where $\underline{c}_X = \min\{X, C_Y\}$. It follows that for $X \geq C_Y$, $G(X) = -\Psi'(X) = 0$. For $X < C_Y$,

$$\begin{aligned}
\Psi'(X) &= -\frac{1-q_0}{r+\gamma} K \left(e^{-\mu X} - \mu X e^{-\mu X}\right) \\
-\Psi'(X) &= \frac{1-q_0}{r+\gamma} K e^{-\mu X} \left(1 - \mu X\right).
\end{aligned}$$

Thus, $G(X) = -\Psi'(X)$ is continuous and decreasing in X. In the case when $K/\mu e \leq q_0/(1-q_0)$, $G(X)$ is continuous on $[0, \infty)$ and decreasing on $[0, C_H)$ where $C_H = C_Y = 1/\mu$ in this case.

Case 2: When $X \geq c^-$
The only relevant case is when $K/\mu e > q_0/(1-q_0)$ in which case c^- is finite.

$$\begin{aligned}
\Psi(X) &= \frac{q_0\gamma}{r(r+\gamma)} e^{-rD(X)} \\
\Psi'(X) &= -\frac{q_0\gamma}{r(r+\gamma)} r\, e^{-rD(X)} D'(X) = -r\Psi(X)D'(X) \\
G(X) &= \frac{q_0\gamma}{r(r+\gamma)} r\, e^{-rD(X)} D'(X) \\
G'(X) &= \frac{q_0\gamma}{r(r+\gamma)} r\, e^{-rD(X)} \left(D''(X) - r(D'(X))^2\right) \\
G'(X) &= -r^2 \frac{q_0\gamma}{r(r+\gamma)} e^{-rD(X)} (D'(X))^2 + \frac{q_0\gamma}{r+\gamma} e^{-rD(X)} D''(X)
\end{aligned}$$

Thus, $G'(X) < 0$ if $D''(X) < 0$. We consider the following sub-cases:

Case 2a: $c^- < X \leq C_Y$

In this case, $\underline{c}_X = \overline{c}_X = X$. Thus,

$$\begin{aligned}
D(X) &= \frac{\mu}{2\gamma K}\left[(1+K\overline{c}_X)^2 - (1+K\underline{c}_X)^2\right] - \frac{1+K\overline{c}_X}{\gamma}\log\left(\frac{w(\overline{c}_X)}{\mu\underline{c}_X}\right)\\
D(X) &= -\frac{1+KX}{\gamma}\log\left(\frac{w(X)}{\mu X}\right)\\
D'(X) &= -\frac{1}{\gamma}\left[K\log\left(\frac{w(X)}{\mu X}\right) + (1+KX)\frac{\mu X}{w(X)}\left(\frac{w'(X)}{\mu X} - \frac{w(X)}{\mu X^2}\right)\right]\\
&= -\frac{1}{\gamma}\left[K\log\left(\frac{w(X)}{\mu X}\right) + (1+KX)\left(\frac{w'(X)}{w(X)} - \frac{1}{X}\right)\right]\\
&= -\frac{1}{\gamma}\left[K\log\left(\frac{w(X)}{\mu X}\right) + (1+KX)\left(\frac{\left(q'(X)\left(X+\frac{1}{K}\right)+q(X)\right)}{q(X)\left(X+\frac{1}{K}\right)} - \frac{1}{X}\right)\right]\\
&= -\frac{1}{\gamma}\left[K\log\left(\frac{w(X)}{\mu X}\right) + (1+KX)\left(\mu(1-q(X)) - \frac{1}{X(KX+1)}\right)\right].
\end{aligned}$$

Differentiating it again,

$$\begin{aligned}
D''(X) &= -\frac{1}{\gamma}\left[2K\left(\mu(1-q(X)) - \frac{1}{X(KX+1)}\right) + (1+KX)\left(-\mu q'(X) + \frac{2KX+1}{(KX^2+X)^2}\right)\right]\\
&= -\frac{1}{\gamma}\left[\frac{1}{X^2(1+KX)} + \mu K(1-q(X))(2-w(X))\right].
\end{aligned}$$

From the definition of $w(\cdot)$, for $c^- < X \le C_Y$, $w(X) < 1$. Thus, $D''(X) < 0$ which implies that $G'(X) < 0$.

Case 2b: $C_Y < X < C_H$

In this case, $\underline{c}_X = C_Y$, $\overline{c}_X = X$ and $\mu C_Y = 1$. Thus,

$$\begin{aligned}
D(X) &= \frac{\mu}{2\gamma K}\left[(1+KX)^2 - (1+KC_Y)^2\right] - \frac{1+KX}{\gamma}\log\left(w(X)\right)\\
D'(X) &= \frac{\mu}{\gamma}(1+KX) - \frac{K}{\gamma}\log\left(w(X)\right) - \frac{1+KX}{\gamma}\frac{w'(X)}{w(X)}\\
D'(X) &= \frac{\mu}{\gamma}(1+KX) - \frac{K}{\gamma}\log\left(w(X)\right) - \frac{1+KX}{\gamma}\frac{\left(q'(X)\left(X+\frac{1}{K}\right)+q(X)\right)}{q(X)\left(X+\frac{1}{K}\right)}\\
D'(X) &= \frac{\mu}{\gamma}(1+KX) - \frac{K}{\gamma}\log\left(w(X)\right) - \frac{1+KX}{\gamma}\left(\mu(1-q(X)) + \frac{K}{1+KX}\right)\\
D'(X) &= \frac{1}{\gamma}\left[\mu q(X)(1+KX) - K - K\log(w(X))\right] = \frac{K}{\gamma}\left[w(X) - 1 - \log(w(X))\right].
\end{aligned}$$

Differentiating it again,

$$D''(X) = \frac{K}{\gamma} w'(X) \left(1 - \frac{1}{w(X)}\right).$$

Because $q(X)$ is strictly increasing in X, $w'(X) = \mu q'(X)(X + 1/K) + \mu q(X) > 0$. Also, by the definition of w, $w(X) < 1$ for $C_Y < X < C_H$. Therefore, $D''(X) < 0$.

Case 2c: $X \geq C_H$
In this case, $\underline{c}_X = C_Y$ and $\overline{c}_X = C_H$. Thus, both $\Psi(X)$ and $D(X)$ are constant. Therefore, $G(X) = -\Psi'(X) = 0$ in this case.

In all three cases, both $D(X)$ and $D'(X)$ are continuous in X. Therefore, $G(X) = \frac{q_0 \gamma}{r+\gamma} e^{-rD(X)} D'(X)$ is continuous in X on $[0, \infty)$. Hence, proven. □

Proof of Proposition 3. Let $\rho = r + \lambda$. From Lemma 2, for any cumulative exposure X_t given by the contract, $\mathcal{C}^p$. the adversarial firm's distribution over capacity in any PBE is $H^*_{X_t}$ and the government's optimization problem is

$$\max_{x_t^p \in [\phi, 1]} \int_0^\infty e^{-\rho t} \left[x_t^p + \lambda \Psi(X_t)\right] dt,$$

with $\dot{X}_t = x_t^p$. Therefore, the HJB is

$$\rho V(X) = \max_{x \in [\phi, 1]} \left\{x + \lambda \Psi(X) + x V'(X)\right\}.$$

The maximand is linear in x. Hence, the optimal choice is $x_t^{p*} = 1$ when $1 + V'(X) \geq 0$, and $x_t^{p*} = \phi$ when $1 + V'(X) \leq 0$. At equality, both exposure levels are optimal. In line with our convention, right continuity selects full exposure.

From (10), the marginal security cost of any cumulative exposure X is $G(X) = -\Psi'(X)$. From Lemma 3, $G(X)$ is continuous on $[0, \infty)$, strictly decreasing on $[0, C_H)$ and equal to zero on $[C_H, \infty)$. An implication of Lemma 3 is that $G(X)$ is strictly positive on $X \in [0, C_H)$.

Because $G(X) = -\Psi'(X) = 0$ for $X \geq C_H$, $\Psi(X)$ is constant for $X \geq C_H$. Thus, the government chooses $x_t^p = 1$ when the cumulative exposure reaches C_H. Its continuation

value for $X \geq C_H$ is

$$V(X) = \int_0^\infty e^{-\rho t}[1 + \lambda\Psi(C_H)]\,dt = \frac{1 + \lambda\Psi(C_H)}{\rho}. \tag{42}$$

Thus, there exists a cumulative exposure $\widetilde{X} \in [0, C_H]$ such that the government chooses full exposure for all $X \geq \widetilde{X}$. The continuation value at any such X is

$$\begin{aligned} V(X) &= \int_0^{C_H - X} e^{-\rho t}[1 + \lambda\Psi(X + t)]\,dt + e^{-\rho(C_H - X)}V(C_H) \\ &= \int_X^{C_H} e^{-\rho(s-X)}[1 + \lambda\Psi(s)]\,ds + e^{-\rho(C_H - X)}V(C_H) \end{aligned}$$

Integrating by parts,

$$\begin{aligned} V(X) &= -\left[[1 + \lambda\Psi(s)]\frac{1}{\rho}e^{-\rho(s-X)}\right]_X^{C_H} - \int_X^{C_H} \frac{\lambda}{\rho}e^{-\rho(s-X)}G(s)\,ds + e^{-\rho(C_H-X)}V(C_H) \\ &= \frac{1 + \lambda\Psi(X)}{\rho} - \frac{1 + \lambda\Psi(C_H)}{\rho}e^{-\rho(C_H-X)} - \int_X^{C_H} \frac{\lambda}{\rho}e^{-\rho(s-X)}G(s)\,ds + e^{-\rho(C_H-X)}\frac{1 + \lambda\Psi(C_H)}{\rho} \\ &= \frac{1 + \lambda\Psi(X)}{\rho} - \int_X^{C_H} \frac{\lambda}{\rho}e^{-\rho(s-X)}G(s)\,ds. \end{aligned}$$

Differentiating this,

$$\begin{aligned} V'(X) &= -\frac{\lambda}{\rho}G(X) - \frac{\lambda}{\rho}\left[-G(X) + \rho\int_X^{C_H} e^{-\rho(s-X)}G(s)\,ds\right] \\ V'(X) &= -\lambda\int_X^{C_H} e^{-\rho(s-X)}G(s)\,ds. \end{aligned} \tag{43}$$

Since G is strictly decreasing on $[X, C_H]$, $-V'(X)$ is strictly decreasing in X. Because the exposure switches to 1 at $\widetilde{X}$, from the bang–bang policy, it must be that

$$-V'(\widetilde{X}) = \lambda\int_{\widetilde{X}}^{C_H} e^{-\rho(s-\widetilde{X})}G(s)\,ds = 1. \tag{44}$$

Moreover, because $G(X)$ is strictly decreasing in X, if $\lambda\int_0^{C_H} e^{-\rho s}G(s)\,ds \leq 1$, then the optimal exposure is full exposure from $t = 0$.

Next, suppose consider $\lambda\int_0^{C_H} e^{-\rho s}G(s)\,ds > 1$ so that $\widetilde{X} > 0$. Consider a cumulative exposure $\underline{X} < \widetilde{X}$ such that the optimal exposure is ϕ for any $X \in [\underline{X}, \widetilde{X})$. The continuation

value for such an X is

$$V(X) = \int_0^{(\widetilde{X}-X)/\phi} e^{-\rho t}\big[\phi + \lambda\Psi(X+\phi t)\big]\,dt + e^{-\frac{\rho}{\phi}(\widetilde{X}-X)}V(\widetilde{X}).$$

For X in $[\underline{X}, \widetilde{X})$, the HJB satisfies

$$\rho V(X) = \phi + \lambda\Psi(X) + \phi V'(X).$$

Differentiating it,

$$\begin{aligned}
\rho V'(X) &= \lambda\Psi'(X) + \phi V''(X) \\
\rho(1+V'(X)) - \rho &= -\lambda G(X) + \phi V''(X) \\
\phi V''(X) - \frac{\rho}{\phi}(1+V'(X)) &= \lambda G(X) - \rho \\
e^{-\rho X/\phi}\left[\phi V''(X) - \frac{\rho}{\phi}(1+V'(X))\right] &= e^{-\rho X/\phi}[\lambda G(X) - \rho] \\
\frac{d}{dX}\left[e^{-\rho X/\phi}(1+V'(X))\right] &= \frac{1}{\phi}e^{-\rho X/\phi}[\lambda G(X) - \rho]
\end{aligned}$$

By the smooth pasting condition, $V'(X)$ must be continuous at $\widetilde{X}$. Thus, because $1 + V'(\widetilde{X}) = 0$, integrating from X to $\widetilde{X}$ yields

$$\begin{aligned}
0 - e^{-\rho X/\phi}(1+V'(X)) &= \int_X^{\widetilde{X}} \frac{1}{\phi}e^{-\rho s/\phi}[\lambda G(s) - \rho]\,ds \\
1 + V'(X) &= -\frac{1}{\phi}\int_X^{\widetilde{X}} e^{-\rho\frac{s-X}{\phi}}[\lambda G(s) - \rho]\,ds. \qquad (45)
\end{aligned}$$

Consider the R.H.S. Because G is decreasing in X,

$$\begin{aligned}
\int_X^{\widetilde{X}} e^{-\rho\frac{s-X}{\phi}} G(s)\,ds &> \int_X^{\widetilde{X}} e^{-\rho\frac{s-X}{\phi}} G(\widetilde{X})\,ds \\
\frac{\lambda}{\phi}\int_X^{\widetilde{X}} e^{-\rho\frac{s-X}{\phi}} G(s)\,ds &> \frac{\lambda}{\rho}G(\widetilde{X})\left[1 - e^{-\rho\frac{\widetilde{X}-X}{\phi}}\right]. \qquad (46)
\end{aligned}$$

From (44) and (43),

$$1 = \lambda \int_{\widetilde{X}}^{C_H} e^{-\rho \frac{s-X}{\phi}} G(s)\, ds < \lambda G(\widetilde{X}) \int_{\widetilde{X}}^{\infty} e^{-\rho(s-\widetilde{X})}\, ds = \frac{\lambda}{\rho} G(\widetilde{X}). \tag{47}$$

From (46) and (47),

$$\begin{aligned} \frac{\lambda}{\phi} \int_{X}^{\widetilde{X}} e^{-\rho \frac{s-X}{\phi}} G(s)\, ds > \left[1 - e^{-\rho \frac{\widetilde{X}-X}{\phi}}\right] &= \frac{\rho}{\phi} \int_{X}^{\widetilde{X}} e^{-\rho \frac{s-X}{\phi}}\, ds \\ 0 > -\frac{1}{\phi} \int_{X}^{\widetilde{X}} e^{-\rho \frac{s-X}{\phi}} [\lambda G(s) - \rho]\, ds. & \end{aligned} \tag{48}$$

This implies that $1 + V'(X) < 0$ for $X \in [\underline{X}, \widetilde{X})$. However, (46) holds for any $X < \widetilde{X}$. Thus, it must be that $\underline{X} = 0$. Therefore, the optimal exposure path is $x_t^{p*} = \phi$ until the cumulative exposure reaches $\widetilde{X}$ and $x_t^{p*} = 1$ thereafter. With minimum stakes, the cumulative exposure reaches $\widetilde{X}$ at $\widetilde{X}/\phi$. Therefore, the optimal exposure path is $x_t^{p*} = \phi$ for $t < \widetilde{\tau}^* = \widetilde{X}/\phi$ and $x_t^{p*} = 1$ for $t \geq \widetilde{\tau}^*$.

This completes the proof.

□

Proof of Lemma 4. Fix $X \in [\phi T, T]$ and consider any feasible exposure path x^p with $X_T = X$. At every $t \in [0, T]$, because $x_t^p \in [\phi, 1]$, $X_t = \int_0^t x_s^p\, ds \leq t$ and $X - X_t = \int_t^T x_s^p\, ds \geq \phi(T - t)$. Therefore, at every $t \in [0, T]$,

$$X_t \leq \min\{t, X - \phi(T - t)\}. \tag{49}$$

Consider a front-loaded path $x^B(X)$ with switching time $\tau(X)$, where exposure switches from 1 to ϕ. The switching time satisfies

$$\begin{aligned} \tau(X) + \phi(T - \tau(X)) &= X \\ \tau(X) &= \frac{X - \phi T}{1 - \phi}. \end{aligned} \tag{50}$$

Therefore, at every t, the cumulative exposure under this front-loaded exposure path is $X_t^B = \min\{t, X - \phi(T - t)\}$.

The economic gain for the government under any exposure path x^p until T is

$$\int_0^T e^{-rt} x_t^p \, dt = e^{-rT} X_T + r \int_0^T e^{-rt} X_t \, dt = e^{-rT} X + r \int_0^T e^{-rt} X_t \, dt. \tag{51}$$

Since $e^{-rT}X$ is fixed, maximizing $\int_0^T e^{-rt} x_t^p \, dt$ is equivalent to maximizing $\int_0^T e^{-rt} X_t \, dt$. At every t, $X_t \leq \min\{t, X - \phi(T-t)\}$ (from (49)) and $X_t^B = \min\{t, X - \phi(T-t)\}$. Therefore,

$$\int_0^T e^{-rt} X_t \, dt \;\leq\; \int_0^T e^{-rt} X_t^B \, dt,$$

with strict inequality unless $X_t = X_t^B$ a.e. Therefore, $x^B(X)$ maximizes the government's economic gains given in equation (51). Hence, proven.

□

Proof of Proposition 4. Fix the cumulative exposure when risk begin at time T at X_T. From Lemma 2, every PBE induces $H_t = H^*_{X_t}$ for $t \geq T$. Feasible capacity processes are continuous in their distribution, and X_t and H^*_X are continuous in t and X respectively. Therefore, at $t = T$, it must be that in any PBE, $H_T = H^*_{X_T}$.

Because hostility cannot arrive before T, all pre-T exposure paths that produce the same X_T, generate the same continuation value for the government at T. Denote the continuation value for the government at T by $V(X_T)$. From Lemma 4, discounting results in the government front loading X_T so, conditional on X_T, the pre-T exposure path is

$$x_t^p = \begin{cases} 1, & t < \tau(X_T), \\ \phi, & \tau(X_T) \leq t < T, \end{cases} \qquad \tau(X_T) = \frac{X_T - \phi T}{1 - \phi}. \tag{52}$$

Thus, pre-T, the government's problem is to choose $X_T \in [\phi T, T]$ to maximize

$$W_A(X_T, T) = \phi \frac{1 - e^{-rT}}{r} + (1 - \phi) \frac{1 - e^{-r\tau(X_T)}}{r} + e^{-rT} V(X_T). \tag{53}$$

Let X_T^* be an optimal cumulative exposure at T. When X_T^* lies in the interior, i.e., $X_T^* =$

$X^I(T) \in (\phi T, T)$, by the FOC, it satisfies

$$0 = e^{-r\tau(X^I(T)} + e^{-rT} V'(X^I(T))$$
$$V'(X^I(T)) = -\exp\left(\frac{r(T - X^I(T))}{1-\phi}\right). \tag{54}$$

If $X_T^* = \phi T$, then it must be that $-V'(\phi T) \geq e^{rT}$, and $X_T^* = T$ implies $-V'(T) \leq 1$.

Because the government's problem for $t \geq T$ is the same as the persistent-risk problem, from equation (9), the post-T HJB is

$$(r+\lambda)V(X) = \max_{x \in [\phi, 1]} \{x + \lambda\Psi(X) + xV'(X)\}.$$

Let $\widetilde{X} = \phi\widetilde{\tau}^*$, where $\widetilde{\tau}^*$ is characterized in Proposition 3. Then, from Proposition 3, the post-T exposure path is,

$$x^*(X) = \begin{cases} \phi, & X < \widetilde{X}, \\ 1, & X \geq \widetilde{X}. \end{cases} \tag{55}$$

Equations (52) and (55) show that for any X_T^*, there can be at most two cutoffs τ_1^* and τ_2^* such that the government's optimal exposure goes from 1 to ϕ at τ_1^* and then back to 1 at τ_2^*. The two cutoffs are

$$\tau_1^* = \frac{X_T^* - \phi T}{1-\phi}, \qquad \tau_2^* = T + \frac{\max\{0, \widetilde{X} - X_T^*\}}{\phi}. \tag{56}$$

Next, we characterize X_T^*. Let $Q(X) = -V'(X)$ and recall that $G(X) = -\Psi'(X)$. Because $x^*(X) = 1$ for $X \geq \widetilde{X}$ (from (55)), the HJB becomes

$$(r+\lambda)V(X) = 1 + \lambda\Psi(X) + V'(X).$$

Differentitating it gives,

$$Q'(X) = (r+\lambda)Q(X) - \lambda G(X).$$

Because $G(X) = -\Psi'(X) = 0$ for $X \geq C_H$ (Lemma 3), $\Psi(X)$ is constant for $X \geq C_H$. Thus, the government chooses full exposure from then on. Its continuation value, from (42), is

$$V(X) = \frac{1 + \lambda\Psi(C_H)}{r+\lambda}. \tag{57}$$

Therefore, $Q(C_H) = -V'(C_H) = 0$. We proceed using the same approach as in the proof of Proposition 3. Solving backward yields,

$$Q(X) = \lambda \int_X^{C_H} e^{-(r+\lambda)(s-X)} G(s)\, ds, \tag{58}$$

for $X \geq \widetilde{X}$. If $\widetilde{X} > 0$, at $X = \widetilde{X}$, from (44)

$$Q(\widetilde{X}) = \lambda \int_{\widetilde{X}}^{C_H} e^{-(r+\lambda)(s-\widetilde{X})} G(s)\, ds = 1,$$

which is the cutoff equation from Proposition 3. If $\widetilde{X} \leq X_T^*$, the government chooses full exposure for all $t \geq T$.

If $\widetilde{X} > X_T^*$, the government chooses minimum exposure ϕ after T until cumulative exposure reaches $\widetilde{X}$, so for $X_T^* < X < \widetilde{X}$, the HJB is

$$(r+\lambda)V(X) = \phi + \lambda\Psi(X) + \phi V'(X).$$

Again, differentiating it yields the ODE,

$$\phi Q'(X) = (r+\lambda)Q(X) - \lambda G(X). \tag{59}$$

Using the same approach as in the proof of Proposition 3, we solve this ODE backwards from $Q(\widetilde{X}) = 1$. From the expression for $1 + V'(X)$ in (45), and using $Q(X) = -V'(X)$ yields

$$\begin{aligned}
1 - Q(X) &= -\frac{1}{\phi} \int_X^{\widetilde{X}} e^{-(r+\lambda)\frac{s-X}{\phi}} [\lambda G(s) - (r+\lambda)]\, ds \\
Q(X) &= \frac{1}{\phi} \int_X^{\widetilde{X}} e^{-(r+\lambda)\frac{s-X}{\phi}} [\lambda G(s) - (r+\lambda)]\, ds + 1 \\
Q(X) &= \frac{\lambda}{\phi} \int_X^{\widetilde{X}} e^{-(r+\lambda)\frac{s-X}{\phi}} G(s)\, ds + e^{-\frac{r+\lambda}{\phi}(\widetilde{X}-X)}.
\end{aligned} \tag{60}$$

Now, $Q(X) > 1 \iff X < \widetilde{X}$. Because $X^I(T) < T$, from (54), $Q(X^I(T)) = \exp\left(\frac{r(T-X^I(T))}{1-\phi}\right) >$

1. Thus, $X^I(T) < \widetilde{X}$. Combining this with (60),

$$e^{-\frac{r+\lambda}{\phi}(\widetilde{X}-X^I(T))} + \frac{\lambda}{\phi}\int_{X^I(T)}^{\widetilde{X}} e^{-\frac{(r+\lambda)}{\phi}(s-X^I(T))}G(s)\,ds = \exp\left(\frac{r(T-X^I(T))}{1-\phi}\right). \tag{61}$$

Next, we show that there can be at most one local interior maximum $X^I(T)$. We begin by showing that Q is strictly decreasing below C_H. From (47), $\lambda G(\widetilde{X}) > r+\lambda$. Because G is decreasing (Lemma 3), for $X < \widetilde{X}$, $\frac{\lambda G(X)}{r+\lambda} > 1$. Using the expression for $Q(X)$,

$$\begin{aligned} Q(X) &= e^{-\frac{r+\lambda}{\phi}(\widetilde{X}-X)} + \frac{\lambda}{\phi}\int_{X}^{\widetilde{X}} e^{-(r+\lambda)\frac{s-X}{\phi}}G(s)\,ds \\ &< e^{-\frac{r+\lambda}{\phi}(\widetilde{X}-X)} + \frac{\lambda G(X)}{r+\lambda}\left[1-e^{-\frac{r+\lambda}{\phi}(\widetilde{X}-X)}\right] < \frac{\lambda G(X)}{r+\lambda}. \end{aligned} \tag{62}$$

The first inequality follows from $G(s) < G(X)$ for $s > X$; the second follows from $\lambda G(X)/(r+\lambda) > 1$. Therefore, the ODE for minimum exposure gives

$$\phi Q'(X) = (r+\lambda)Q(X) - \lambda G(X) < 0.$$

For $\widetilde{X} < X < C_H$, from (47), we get

$$Q(X) = \lambda\int_{X}^{C_H} e^{-(r+\lambda)(s-X)}G(s)\,ds < \frac{\lambda G(X)}{r+\lambda},$$

which implies that $Q'(X) = (r+\lambda)Q(X) - \lambda G(X) < 0$. Since Q is continuous at $\widetilde{X}$, it is strictly decreasing on $[0, C_H)$, and equals zero for $X \geq C_H$.

Next, from (53),

$$\begin{aligned} W_A(X,T) &= \phi\frac{1-e^{-rT}}{r} + (1-\phi)\frac{1-e^{-r\tau(X)}}{r} + e^{-rT}V(X) \\ \frac{\partial W_A(X,T)}{\partial X} &= e^{-r\tau(X)} - e^{-rT}Q(X). \end{aligned} \tag{63}$$

Thus, at any interior critical point, $e^{-r\tau(X)} = e^{-rT}Q(X)$. At this critical point,

$$\begin{aligned}
\frac{\partial^2 W_A(X,T)}{\partial X^2} &= -\frac{r}{1-\phi}e^{-r\tau(X)} - e^{-rT}Q'(X) \\
\frac{\partial^2 W_A(X,T)}{\partial X^2} &= -\frac{r}{1-\phi}e^{-rT}Q(X) - e^{-rT}Q'(X) \\
\frac{\partial^2 W_A(X,T)}{\partial X^2} &= -\frac{e^{-rT}}{\phi(1-\phi)}\left((r+\lambda(1-\phi))Q(X) - \lambda(1-\phi)G(X)\right), \qquad (64)
\end{aligned}$$

where the last equality follows because $X^I(T) < \widetilde{X}$ and we use the HJB $\phi Q'(X) = (r + \lambda)Q(X) - \lambda G(X)$. Define $E(X) := (r + \lambda(1-\phi))Q(X) - \lambda(1-\phi)G(X)$. Thus,

$$\frac{\partial^2 W_A(X,T)}{\partial X^2} = -\frac{e^{-rT}}{\phi(1-\phi)}E(X).$$

Therefore, a necessary SOC for an interior local maximum is $E(X) \geq 0$.

At $X = \widetilde{X}$, $Q(\widetilde{X}) = 1$ and $G(\widetilde{X}) > 1 + \frac{r}{\lambda}$ so $1 - \frac{r}{\lambda[G(\widetilde{X})-1]} > 0$. Assumption 1 implies that $\phi \leq 1 - \frac{r}{\lambda[G(\widetilde{X})-1]}$. Thus,

$$E(\widetilde{X}) = r + \lambda(1-\phi) - \lambda(1-\phi)G(\widetilde{X}) = r - \lambda(1-\phi)[G(\widetilde{X}) - 1] \leq 0.$$

For $X < \widetilde{X}$, differentiating $E(X)$, we get

$$\begin{aligned}
E'(X) &= \big[r + \lambda(1-\phi)\big]Q'(X) - \lambda(1-\phi)G'(X) \\
E'(X) &= \frac{r+\lambda}{\phi}E(X) - \lambda\big[rG(X) + (1-\phi)G'(X)\big] \\
E'(X) - \frac{r+\lambda}{\phi}E(X) &= -\lambda\big[rG(X) + (1-\phi)G'(X)\big] \\
\frac{d}{dX}\left[e^{-\frac{r+\lambda}{\phi}X}E(X)\right] &= -\lambda e^{-\frac{r+\lambda}{\phi}X}\big[rG(X) + (1-\phi)G'(X)\big]. \qquad (65)
\end{aligned}$$

We then determine the sign of $rG(X) + (1-\phi)G'(X)$. From the proof of Lemma 3, whenever the government offers full exposure during hostility,

$$\begin{aligned}
G(X) &= \frac{(1-q_0)K}{r+\gamma}e^{-\mu X}(1-\mu X) \\
G'(X) &= -\frac{(1-q_0)K}{r+\gamma}\mu e^{-\mu X}(2-\mu X).
\end{aligned}$$

Therefore,

$$\frac{rG(X)+(1-\phi)G'(X)}{G(X)} = r-(1-\phi)\mu\frac{2-\mu X}{1-\mu X}.$$

The right-hand side is strictly decreasing in X. Since $G(X)>0$ for $X<\widetilde{X}$, $rG(X)+(1-\phi)G'(X)$ has the same sign as $r-(1-\phi)\mu\frac{2-\mu X}{1-\mu X}$. Thus, $rG(X)+(1-\phi)G'(X)$ can change sign at most once and only from positive to negative.

When the government offers a screening contract during hostility, from the proof of Lemma 3, $G(X)=r\Psi(X)D'(X)$, $D'(X)>0$, $D''(X)<0$, and $0<\Psi(X)\leq\frac{q_0\gamma}{r(r+\gamma)}$. Therefore,

$$\frac{G'(X)}{G(X)} = \frac{\Psi'(X)}{\Psi(X)}+\frac{D''(X)}{D'(X)} = -\frac{G(X)}{\Psi(X)}+\frac{D''(X)}{D'(X)} < -\frac{G(X)}{\Psi(X)}. \tag{66}$$

From (62), $G(X)>\frac{r+\lambda}{\lambda}Q(X)>\frac{r+\lambda}{\lambda}$. Combining these inequalities,

$$\begin{aligned}\frac{rG(X)+(1-\phi)G'(X)}{G(X)} &= r+(1-\phi)\frac{G'(X)}{G(X)}\\ &< r-(1-\phi)\frac{G(X)}{\Psi(X)}\\ &< r-(1-\phi)\frac{(r+\lambda)r(r+\gamma)}{\lambda q_0\gamma} \leq -\frac{r^2}{\lambda}<0,\end{aligned}$$

where the last inequality follows because $\phi\leq 1-\frac{q_0\gamma}{r+\gamma}$ from Assumption 1. Since $G(X)>0$, $rG(X)+(1-\phi)G'(X)<0$ in the case when the government offers a screening contract during hostility.

Combining this with the case when government offers full exposure during hostility, the expression $rG(X)+(1-\phi)G'(X)$ can change sign at most once and go from positive to negative.

From (65),

$$\frac{d}{dX}\left[e^{-\frac{r+\lambda}{\phi}X}E(X)\right] = -\lambda e^{-\frac{r+\lambda}{\phi}X}\left[rG(X)+(1-\phi)G'(X)\right].$$

Since $-\lambda e^{-\frac{r+\lambda}{\phi}X}$ is negative, the LHS can change sign at most once from negative to positive. Therefore, it follows that $e^{-\frac{r+\lambda}{\phi}X}E(X)$ can be either monotonic (increasing or decreasing) or first decreases and then increases. Because $e^{-\frac{r+\lambda}{\phi}X}>0$, $E(X)$ has the same sign as $e^{-\frac{r+\lambda}{\phi}X}E(X)$. Thus, since $E(\widetilde{X})\leq 0$, the increasing terminal portion must be non-positive

throughout ($\because X < \widetilde{X}$). On the preceding decreasing portion, $E(X)$ can change sign at most once, from positive to negative. Hence, $E(X) > 0$ only on an initial interval if such an interval exists.

Next, coming back to the government's objective, from (53),

$$\begin{aligned}
\frac{\partial W_A(X,T)}{\partial X} &= e^{-r\tau(X)} - e^{-rT}Q(X) \\
\frac{\partial^2 W_A(X,T)}{\partial X^2} &= -\frac{r}{1-\phi}e^{-r\tau(X)} - e^{-rT}Q'(X) \\
\frac{\partial^2 W_A(X,T)}{\partial X^2} &= -\frac{r}{1-\phi}\left(\frac{\partial W_A(X,T)}{\partial X} + e^{-rT}Q(X)\right) - e^{-rT}Q'(X) \\
\frac{\partial^2 W_A(X,T)}{\partial X^2} + \frac{r}{1-\phi}\frac{\partial W_A(X,T)}{\partial X} &= -e^{-rT}\left(\frac{r}{1-\phi}Q(X) + Q'(X)\right), \qquad (67)
\end{aligned}$$

where we get the last equality from substituting $e^{-r\tau(X)}$ from the equation for $\frac{\partial W_A(X,T)}{\partial X}$. Because $X < \widetilde{X}$, the minimum exposure ODE for Q gives

$$\begin{aligned}
\phi Q'(X) &= (r+\lambda)Q(X) - \lambda G(X) \\
Q'(X) &= \frac{(r+\lambda)}{\phi}Q(X) - \frac{\lambda}{\phi}G(X) \qquad (68) \\
\frac{r}{1-\phi}Q(X) + Q'(X) &= \frac{r}{1-\phi}Q(X) + \frac{(r+\lambda)}{\phi}Q(X) - \frac{\lambda}{\phi}G(X) \\
\frac{r}{1-\phi}Q(X) + Q'(X) &= \frac{r\phi Q(X) + (r+\lambda)(1-\phi)Q(X) - \lambda(1-\phi)G(X)}{\phi(1-\phi)} \\
\frac{r}{1-\phi}Q(X) + Q'(X) &= \frac{[r\phi + r(1-\phi) + \lambda(1-\phi)]Q(X) - \lambda(1-\phi)G(X)}{\phi(1-\phi)} \\
\frac{r}{1-\phi}Q(X) + Q'(X) &= \frac{E(X)}{\phi(1-\phi)}. \qquad (69)
\end{aligned}$$

Substituting this in (67),

$$\frac{\partial^2 W_A(X,T)}{\partial X^2} + \frac{r}{1-\phi}\frac{\partial W_A(X,T)}{\partial X} = -e^{-rT}\frac{E(X)}{\phi(1-\phi)}.$$

Multiplying both sides by $e^{\frac{rX}{1-\phi}}$,

$$\frac{d}{dX}\left[e^{\frac{rX}{1-\phi}}\frac{\partial W_A(X,T)}{\partial X}\right] = -\frac{e^{-r\left(T-\frac{X}{1-\phi}\right)}}{\phi(1-\phi)}E(X). \qquad (70)$$

Since $E(X) > 0$ only on an initial interval, $\frac{\partial W_A(X,T)}{\partial X}$ can change sign from positive to negative at most once. Thus, $\frac{\partial W_A(X,T)}{\partial X}$ can have at most one solution that is a local maximum. Thus, $W_A(X,T)$ has at most one interior local maximum $X^I(T)$ whenever it exists and is feasible.

Therefore, the government's global optimum cumulative exposure must belong to the set $\{\phi T, X^I(T), T\}$. To see how $X^I(T)$ changes with T, from (54),

$$\log Q(X^I(T)) = \frac{r[T - X^I(T)]}{1-\phi}. \tag{71}$$

Differentiating w.r.t. T,

$$\begin{aligned}
\frac{Q'(X^I(T))}{Q(X^I(T))}\frac{dX^I(T)}{dT} &= \frac{r}{1-\phi}\left[1 - \frac{dX^I(T)}{dT}\right] \\
\frac{dX^I(T)}{dT} &= \frac{(1-\phi)rQ(X^I(T))}{(1-\phi)[rQ(X^I(T)) + (1-\phi)Q'(X^I(T))]} \\
\frac{dX^I(T)}{dT} &= \frac{rQ(X^I(T))}{rQ(X^I(T)) + (1-\phi)Q'(X^I(T))} > 1.
\end{aligned}$$

The last inequality follows because from (68),

$$rQ(X^I(T)) + (1-\phi)Q'(X^I(T)) = \frac{E(X^I(T))}{\phi} > 0,$$

so the denominator is positive and $Q'(X) < 0$ for any $X < C_H$. Therefore, $\frac{d}{dT}\left[X^I(T) - \phi T\right] > 0$ and $\frac{d}{dT}\left[X^I(T) - T\right] > 0$ which implies that both $X^I_T - \phi T$ and $X^I_T - T$ are strictly increasing. Hence, the set of T for which $X^I(T)$ is feasible is an interval.

We next compare the government's payoff for the three candidate exposures. Define

$$W_L(T) = W_A(\phi T, T), \qquad W_I(T) = W_A(X^I(T), T), \qquad W_F(T) = W_A(T, T),$$

where $W_I(T)$ is defined whenever the interior candidate exists and is feasible. First, compare

the interior candidate with minimal exposure. Since $\tau(\phi T) = 0$,

$$\begin{aligned} D_{IL}(T) &= W_I(T) - W_L(T) \\ &= (1-\phi)\frac{1-e^{-r\tau(X^I(T))}}{r} + e^{-rT}V(X^I(T)) - e^{-rT}V(\phi T) \\ &= (1-\phi)\frac{1-e^{-r\tau(X^I(T))}}{r} - e^{-rT}\int_{\phi T}^{X^I(T)} Q(s)\,ds, \end{aligned}$$

where the last equality follows from substituting $Q(X) = -V'(X)$. Differntiating the above equation and using the interior first order condition $e^{-r\tau(X^I(T))} = e^{-rT}Q(X^I(T))$,

$$\begin{aligned} D'_{IL}(T) &= (1-\phi)e^{-r\tau(X^I(T))}\frac{d}{dT}\tau(X^I(T)) + re^{-rT}\int_{\phi T}^{X^I(T)} Q(s)\,ds - e^{-rT}\left[Q(X^I(T))\frac{dX^I(T)}{dT} - \phi Q(\phi T)\right] \\ &= e^{-rT}Q(X^I(T))\left[\frac{dX^I(T)}{dT} - \phi\right] + re^{-rT}\int_{\phi T}^{X^I(T)} Q(s)\,ds - e^{-rT}\left[Q(X^I(T))\frac{dX^I(T)}{dT} - \phi Q(\phi T)\right] \\ &= e^{-rT}\left[r\int_{\phi T}^{X^I(T)} Q(s)\,ds + \phi(Q(\phi T) - Q(X^I(T)))\right] > 0. \end{aligned}$$

The last inequality follows because $Q(X)$ is decreasing in X. Hence, the payoff under $X^I(T)$ can cross that under minimal exposure at most once. Denote this crossing, if it exists, by T^A_{IL}.

Next, compare full exposure with minimal exposure:

$$D_{FL}(T) = W_F(T) - W_L(T) = (1-\phi)\frac{1-e^{-rT}}{r} - e^{-rT}\int_{\phi T}^{T} Q(s)\,ds.$$

At any $T > 0$ such that $D_{FL}(T) = 0$, it must be that

$$\frac{rT}{e^{rT}-1}\frac{1}{1-\phi}\int_{\phi}^{1} Q(uT)\,du = 1. \tag{72}$$

The L.H.S. in (72) is strictly decreasing in T because Q is weakly positive and decreasing, and $\frac{rT}{e^{rT}-1}$ is positive and strictly decreasing for $T > 0$. Thus, the L.H.S. converges to $Q(0) > 1$ as T goes to zero and to zero as $T \to \infty$. Hence, there exists a unique $T^A_{FL} > 0$ such that $W_F(T^A_{FL}) = W_L(T^A_{FL})$.

Finally, compare full exposure with the interior candidate. Define $\widetilde{D}_{FI}(T) := e^{rT}[W_F(T) -$

$W_I(T)]$. Then,

$$\widetilde{D}_{FI}(T) = (1-\phi)\frac{e^{r(T-\tau(X^I(T)))}-1}{r} - \int_{X^I(T)}^{T} Q(s)\,ds.$$

Differentiating and using the FOC for $X^I(T)$

$$\begin{aligned}
\widetilde{D}'_{FI}(T) &= e^{r[T-\tau(X^I(T))]}\left[1-\frac{dX^I(T)}{dT}\right] - Q(T) + Q(X^I(T))\frac{dX^I(T)}{dT} \\
&= Q(X^I(T))\left[1-\frac{dX^I(T)}{dT}\right] - Q(T) + Q(X^I(T))\frac{dX^I(T)}{dT} \\
&= Q(X^I(T)) - Q(T) > 0,
\end{aligned}$$

where the inequality follows because $X^I(T) < T$, $X^I(T) < \widetilde{X}$, and Q is strictly decreasing below C_H and zero thereafter. Thus, full exposure can overtake $X^I(T)$ at most once. Denote this crossing, if it exists, by T^A_{FI}.

Because $X^I(T)$ is feasible over an interval and there can exist at most one of each of these crossings: T^A_{IL}, T^A_{FL}, and T^A_{FI}, we can define two thresholds $0 \leq \underline{T}^A \leq \overline{T}^A$ such that the government sets ϕT for $T < \underline{T}^A$, choose $X^I(T)$ for $\underline{T}^A \leq T < \overline{T}^A$, and set T for $T \geq \overline{T}^A$.

If $T^A_{FL} \leq T^A_{IL}$, it implies that the government chooses full exposure before $X^I(T)$ becomes feasible. Therefore, the government switches directly from ϕ to 1 and $\underline{T}^A = \overline{T}^A = T^A_{FL}$.

If $T^A_{FL} > T^A_{IL}$, $X^I(T)$ becomes feasible while minimum exposure is still preferable to full exposure. At T^A_{IL},

$$W_I(T^A_{IL}) = W_L(T^A_{IL}) > W_F(T^A_{IL}).$$

Since $D'_{IL}(T) > 0$, $X^I(T)$ strictly dominates ϕT thereafter. Thus, $X^I(T)$ is globally optimal until full exposure overtakes it at T^A_{FI}. At T^A_{FI}, the government chooses full exposure over both $X^I(T)$ and ϕT. Thus,

$$\underline{T}^A = \min\{T^A_{IL}, T^A_{FL}\}, \qquad \overline{T}^A = \max\{T^A_{FL}, T^A_{FI}\}. \tag{73}$$

Therefore, the optimal cumulative exposure X^*_T at T is

$$X^*_T = \begin{cases} \phi T, & T < \underline{T}_A, \\ X^I(T), & T \in [\underline{T}_A, \overline{T}_A), \\ T, & T \geq \overline{T}_A. \end{cases} \tag{74}$$

Given X_T^*, the two cutoffs τ_1^* and τ_2^* are given by (56). Thus,

$$\tau_1^* = \begin{cases} 0, & T < \underline{T}^A, \\ \tau^I(T), & \underline{T}^A \le T < \overline{T}^A, \\ T, & T \ge \overline{T}^A, \end{cases}$$

where $\tau^I(T) = \frac{X^I(T) - \phi T}{1-\phi}$. Define $X_T(\tau) = \phi T + (1-\phi)\tau$. Thus, $X_T(\tau^I(T)) = X^I(T)$. Because $X^I(T)$ satisfies (61), $\tau^I(T)$ satisfies

$$e^{-\frac{r+\lambda}{\phi}\left[\widetilde{X} - X_T(\tau^I(T))\right]} + \frac{\lambda}{\phi}\int_{X_T(\tau^I(T))}^{\widetilde{X}} e^{-\frac{r+\lambda}{\phi}\left[s - X_T(\tau^I(T))\right]} G(s)\, ds = e^{r\left(T - \tau^I(T)\right)},$$

We can then write τ_2^* as

$$\tau_2^* = T + \frac{1}{\phi}\max\{0, \widetilde{X} - X_T^*\} = T + \frac{1}{\phi}\max\{0, \widetilde{X} - X_T(\tau_1^*)\}.$$

For an interior optimum, $X^I(T) < \widetilde{X}$, so $\tau_2^* > T$. If $X_T^* = T$, it follows that $T \ge \widetilde{X}$, otherwise the government would choose minimum exposure at T. Hence, $\tau_2^* = T$. Therefore, $0 \le \tau_1^* \le T \le \tau_2^*$, and the optimal exposure path is

$$x_t^{p*} = \begin{cases} 1, & t < \tau_1^*, \\ \phi, & \tau_1^* \le t < \tau_2^*, \\ 1, & t \ge \tau_2^*. \end{cases}$$

Hence, proven.

□

Proof of Proposition 5. Fix all parameters other than T. Whenever the optimal exposure path includes a positive period of restriction, reopening occurs when the cumulative exposure reaches $\widetilde{X}$ (Proposition 3 and Proposition 4), where $\widetilde{X}$ is given by (44), which is independent of T. For a fixed T, denote the two cutoffs for the U-shaped policy under anticipated risk by $\tau_1^*(T)$ and $\tau_2^*(T)$. Thus, from Proposition 4,

$$\begin{aligned} \widetilde{X} &= \tau_1^*(T) + \phi\left[\tau_2^*(T) - \tau_1^*(T)\right] \\ \tau_2^*(T) &= \frac{1}{\phi}\left[\widetilde{X} - (1-\phi)\tau_1^*(T)\right]. \end{aligned} \tag{75}$$

From (75), $\tau_2^*(T)$ is decreasing as $\tau_1^*(T)$ increases. Therefore, it suffices to show that $\tau_1^*(T)$ is weakly increasing in T.

From Proposition 4, when the government restricts exposure from the beginning, $\tau_1^*(T) = 0$ and $\tau_2^*(T) = \widetilde{X}/\phi$. Then, the duration of restriction is constant in T.

When the optimal exposure path is U-shaped, which happens $T_{FL}^A > T_{IL}^A$ and $T \in [\underline{T}^A, \overline{T}^A)$. Then, from Equation (74), $X_T^* = X^I(T)$ and from Lemma 4, $\tau_1^*(T) = \frac{X^I(T) - \phi T}{1-\phi}$. From the calculations following Equation (71),

$$\frac{dX^I(T)}{dT} = \frac{rQ(X^I(T))}{rQ(X^I(T)) + (1-\phi)Q'(X^I(T))} > 1.$$

Differentiating $\tau_1^*(T)$ w.r.t T yields

$$\frac{d\tau_1^*(T)}{dT} = \frac{1}{1-\phi}\left[\frac{dX^I(T)}{dT} - \phi\right] > \frac{1}{1-\phi}\left[1-\phi\right] = 1.$$

Finally, from Proposition 4, the expression for τ_1^* implies that as T increases, the optimal exposure goes from a probationary pattern to a U-shaped pattern as τ_1^* increases from 0 to a positive value. This implies that $\tau_2^*(T)$ cannot increase.

Therefore, as T increases, $\tau_1^*(T)$ weakly increases and $\tau_2^*(T)$ weakly decreases. Hence, proven.

□

Proof of Corollary 1. The proof follows directly from Proposition 4. The threshold for cumulative exposure, $\widetilde{X}$, at which the government reopens after retrenchment remains the same as under persistent risk. Since $x_t^p = 1$ for $t < \tau_1^*$ and ϕ for $t \in [\tau_1^*, \tau_2^*)$,

$$\widetilde{X} = \tau_1^* + \phi(\tau_2^* - \tau_1^*).$$

Consequently, the duration of retrenchment satisfies

$$\tau_2^* - \tau_1^* = \frac{\widetilde{X} - \tau_1^*}{\phi} \leq \frac{\widetilde{X}}{\phi} = \widetilde{\tau}^*,$$

where $\widetilde{\tau}^*$ is the duration of restriction under persistent risk. If no retrenchment occurs, its duration is zero and the inequality holds trivially.

□

Proof of Proposition A.6. The result follows directly from Theorem 1 in Kolb and Madsen (2023) after substituting the effective harm parameter $K' = 1 + KC_{T_h}$ and the beliefs at the start of hostility as q_{T_h}. Theorem 1 (Kolb and Madsen, 2023), gives the optimal contract and the hostility continuation are calculated from the payoffs in the proof of Theorem 1. □

Proof of Proposition B.7. By Lemma 4, for any total exposure $X \in [\phi T_h, T_h]$, the government implements X using the front-loaded bang–bang path $x^B(X)$. Therefore, its objective function is

$$W(X, T_h) = \phi \frac{1 - e^{-rT_h}}{r} + (1 - \phi)\frac{1 - e^{-r\tau(X)}}{r} + e^{-rT_h}\Psi(X),$$

where $\tau(X) = \frac{X - \phi T_h}{1 - \phi}$.

Rest of the proof uses a similar approach as that of the proof of Proposition 4. Let $G(X) = -\Psi'(X)$. From Lemma 3, G is continuous, strictly positive and strictly decreasing on $[0, C_H)$, and equals zero for $X \geq C_H$. Differentiating the government's objective gives

$$\frac{\partial W}{\partial X}(X, T_h) = e^{-r\tau(X)} - e^{-rT_h}G(X). \tag{76}$$

We first consider the case $G(0) = \frac{(1-q_0)K}{r+\gamma} \leq 1$. Because G is decreasing, $G(X) < 1$ for every $X > 0$. Moreover, $X \leq T_h$ implies $\tau(X) \leq T_h$. Therefore,

$$e^{-r\tau(X)} \geq e^{-rT_h} \geq e^{-rT_h}G(X).$$

Hence, $\frac{\partial W}{\partial X}(X, T_h) \geq 0$ throughout the feasible interval. Under our convention that the governemt selects a higher cumulative exposure in case of indifference, $X^*(T_h) = T_h$ for every T_h. This case can be represented by setting $\underline{T}' = \overline{T}' = 0$.

For the remainder of the proof, suppose that $G(0) > 1$. Define

$$\widehat{c} = \begin{cases} 1/\mu, & \dfrac{K}{\mu e} \leq \dfrac{q_0}{1 - q_0}, \\ c^-, & \dfrac{K}{\mu e} > \dfrac{q_0}{1 - q_0}. \end{cases}$$

When the government offers full exposure during hostility, $X \leq \widehat{c}$, Lemma D.6 gives $G(X) =$

$\frac{(1-q_0)K}{r+\gamma}e^{-\mu X}(1-\mu X)$. Hence, an interior critical point satisfies

$$\frac{(1-q_0)K}{r+\gamma}e^{-\mu X}(1-\mu X) = \exp\left(\frac{r(T_h - X)}{1-\phi}\right).$$

Equivalently,

$$\frac{1}{\mu} - X = \frac{r+\gamma}{(1-q_0)\mu K}\exp\left(\frac{rT_h}{1-\phi} + \left[\mu - \frac{r}{1-\phi}\right]X\right). \tag{77}$$

To characterize the solutions of (77), let $R(T_h, X) = \exp\left(-\frac{r(T_h-X)}{1-\phi}\right)G(X)$. Equation (76) can then be written as $\frac{\partial W}{\partial X}(X, T_h) = e^{-r\tau(X)}[1 - R(T_h, X)]$.

Differentiating w.r.t. X,

$$\frac{\partial}{\partial X}\log R(T_h, X) = \frac{r}{1-\phi} - \mu - \frac{1}{1/\mu - X},$$
$$\frac{\partial^2}{\partial X^2}\log R(T_h, X) = -\frac{1}{(1/\mu - X)^2} < 0.$$

Therefore, $\log R(T_h, X)$ is strictly concave in X, so the set $\{X : R(T_h, X) \geq 1\}$ is an interval. Hence, the $R(T_h, X) = 1$ has at most one upward-crossing solution. Because $\frac{\partial W}{\partial X}(X, T_h) = e^{-r\tau(X)}[1 - R(T_h, X)]$, this upward-crossing solution is the unique local maximum. Any other solution is a downward crossing and therefore a local minimum.

Let $X^I(T_h)$ denote this unique upward-crossing solution of (77), whenever it belongs to $[\phi T_h, \min\{T_h, \widehat{c}\}]$. Since $0 \leq X \leq 1/\mu$, $\frac{1}{1/\mu - X} \geq \mu$, and therefore,

$$\frac{\partial}{\partial X}\log R(T_h, X) \leq \frac{r}{1-\phi} - 2\mu.$$

If $r/(1-\phi) \leq 2\mu$, then $R(T_h, X)$ is non-increasing on $[\phi T_h, \min\{T_h, \widehat{c}\}]$ and cannot cross 1 from below. Thus, an upward crossing solution can exist only if $\frac{r}{1-\phi} > 2\mu$. If this inequality does not hold, there is no interior maximum in this case.

Next, suppose that $\frac{K}{\mu e} > \frac{q_0}{1-q_0}$. We show that there can be no interior maximizer for $X \in [c^-, C_H]$. From the proof of Lemma 3, $G(X) = r\Psi(X)D'(X)$, where $D'(X) > 0$, $D''(X) < 0$, and $0 < \Psi(X) \leq \frac{q_0\gamma}{r(r+\gamma)}$. At any interior critical point, (76) implies $G(X) = \exp\left(\frac{r(T_h-X)}{1-\phi}\right) \geq$

1 because $X \leq T_h$. Because G is differentiable within each branch,

$$\frac{G'(X)}{G(X)} = \frac{\Psi'(X)}{\Psi(X)} + \frac{D''(X)}{D'(X)} = -\frac{G(X)}{\Psi(X)} + \frac{D''(X)}{D'(X)} < -\frac{G(X)}{\Psi(X)} < -\frac{1}{\Psi(X)} \leq -\frac{r(r+\gamma)}{q_0\gamma}.$$

By Assumption 1, $\phi \leq 1 - \frac{q_0\gamma}{r+\gamma}$, which is equivalent to $\frac{r}{1-\phi} \leq \frac{r(r+\gamma)}{q_0\gamma}$. Hence, at every interior critical point in $[c^-, C_H]$, $\frac{r}{1-\phi} + \frac{G'(X)}{G(X)} < 0$. Differentiating (76) and applying the first-order condition gives

$$\frac{\partial^2 W}{\partial X^2}(X, T_h) = -\frac{r}{1-\phi} e^{-r\tau(X)} - e^{-rT_h} G'(X) = -e^{-rT_h} G(X) \left[\frac{r}{1-\phi} + \frac{G'(X)}{G(X)}\right] > 0.$$

Thus, every interior critical point on $[c^-, C_H]$ is a strict local minimum. Because G is continuous at the boundaries of the branches (Lemma 3), none of the boundaries create any additional maximum.

It follows that the government's global optimizer must be one of $X_L(T_h) = \phi T_h$, $X_I(T_h)$, or $X_F(T_h) = T_h$, where $X_I(T_h)$ is included only when the upward-crossing interior candidate exists.

Next, $X^I(T_h)$ exists on an interval of T_h. To see this, let $a = \frac{r}{1-\phi}$. Implicitly differentiating the FOC $G(X^I(T_h)) = \exp\left(a[T_h - X^I(T_h)]\right)$ gives

$$\frac{dX^I(T_h)}{dT_h} = \frac{a}{a + G'(X^I(T_h))/G(X^I(T_h))} > 1.$$

The denominator is positive because $X^I(T_h)$ is the upward-crossing solution, and it is smaller than a because $G' < 0$. Therefore, both $X^I(T_h) - \phi T_h$ and $X^I(T_h) - T_h$ are strictly increasing. Hence, the interior candidate can enter and leave the feasible interval at most once.

Let $W_L(T_h) = W(\phi T_h, T_h)$, $W_I(T_h) = W(X^I(T_h), T_h)$, and $W_F(T_h) = W(T_h, T_h)$, where W_I is defined only when $X^I(T_h)$ exists.

Next, we compare these payoffs the same way as we do in the proof of Proposition 4. First, compare the interior candidate with minimal exposure. Let $D_{IL}(T_h) = W_I(T_h) - W_L(T_h)$. Because $\tau(\phi T_h) = 0$,

$$D_{IL}(T_h) = (1-\phi)\frac{1 - e^{-r\tau(X^I(T_h))}}{r} - e^{-rT_h} \int_{\phi T_h}^{X^I(T_h)} G(s)\, ds.$$

Differentiating and applying the first-order condition for $X^I(T_h)$ gives

$$D'_{IL}(T_h) = e^{-rT_h}\left[r\int_{\phi T_h}^{X^I(T_h)} G(s)\,ds + \phi\left\{G(\phi T_h) - G(X^I(T_h))\right\}\right] > 0.$$

Hence, D_{IL} crosses zero at most once. Denote this unique crossing, if it exists, by T'_{IL}.

Next, compare full exposure with minimal exposure. Let $D_{FL}(T_h) = W_F(T_h) - W_L(T_h)$. Because $\tau(T_h) = T_h$,

$$D_{FL}(T_h) = (1-\phi)\frac{1-e^{-rT_h}}{r} - e^{-rT_h}\int_{\phi T_h}^{T_h} G(s)\,ds.$$

The condition $D_{FL}(T_h) = 0$ is equivalent to

$$1 = \frac{rT_h}{e^{rT_h}-1}\frac{1}{1-\phi}\int_{\phi}^{1} G(uT_h)\,du.$$

The first term on the right-hand side is strictly decreasing in T_h. Because G is decreasing, the second term is also weakly decreasing and is strictly decreasing while the interval intersects the region where $G > 0$. Moreover, the right-hand side converges to $G(0) > 1$ as T_h converges to zero and converges to zero as T_h converges to infinity. Therefore, there exists a unique positive T'_{FL} such that

$$W_F(T'_{FL}) = W_L(T'_{FL}).$$

Finally, compare full exposure with the interior candidate. Let

$$D_{FI}(T_h) = W_F(T_h) - W_I(T_h).$$

Multiplying by e^{rT_h} preserves its sign and zeros. Define $\widetilde{D}_{FI}(T_h) = e^{rT_h}D_{FI}(T_h)$. Then,

$$\widetilde{D}_{FI}(T_h) = (1-\phi)\frac{e^{r[T_h-\tau(X^I(T_h))]}-1}{r} - \int_{X^I(T_h)}^{T_h} G(s)\,ds.$$

Differentiating and by FOC, $\widetilde{D}'_{FI}(T_h) = G(X^I(T_h)) - G(T_h) > 0$. Hence, D_{FI} crosses zero at most once. Denote this unique crossing, if it exists, by T'_{FI}.

If $X^I(T_h)$ does not exist, the only candidates are minimal and full exposure. Therefore, $\underline{T}' = \overline{T}' = T'_{FL}$.

Suppose next that $X^I(T_h)$ exists. Consider two cases.

Case 1: $T'_{FL} \leq T'_{IL}$.
For $T_h < T'_{FL}$, ϕT_h dominates both T_h and $X^I(T_h)$. At T'_{FL},

$$W_F(T'_{FL}) = W_L(T'_{FL}) \geq W_I(T'_{FL}).$$

Since $\widetilde{D}_{FI}$ is strictly increasing, $X^I(T_h)$ cannot become more profitable than full exposure. Therefore, the government moves directly from minimal to full exposure at T'_{FL}, and $\underline{T}' = \overline{T}' = T'_{FL}$.

Case 2: $T'_{IL} < T'_{FL}$.
For $T_h < T'_{IL}$, minimal exposure dominates both other candidates. At T'_{IL}, $W_I(T'_{IL}) = W_L(T'_{IL}) > W_F(T'_{IL})$. Hence, $X^I(T_h)$ becomes optimal.

At T'_{FL}, $W_I(T'_{FL}) > W_L(T'_{FL}) = W_F(T'_{FL})$, so the interior candidate remains optimal. Because $\widetilde{D}_{FI}$ is strictly increasing and full exposure is eventually optimal, there exists a unique $T'_{FI} > T'_{FL}$ such that $W_F(T'_{FI}) = W_I(T'_{FI})$. Therefore, $\underline{T}' = T'_{IL}$, and $\overline{T}' = T'_{FI}$.

Combining the cases gives

$$X^*(T_h) = \begin{cases} \phi T_h, & T_h < \underline{T}', \\ X^I(T_h), & T_h \in [\underline{T}', \overline{T}'), \\ T_h, & T_h \geq \overline{T}', \end{cases}$$

where the middle branch is omitted whenever $X^I(T_h)$ does not exist or $\underline{T}' = \overline{T}'$. The higher-exposure tie-breaking rule determines the government's choice at the transition dates.

□

Proof of Proposition C.8. We provide a sketch of the proof here.

The affine reduction implies that G_a is continuous, positive and strictly decreasing before C_H^a, and zero thereafter. Thus, the proof approach remains the same as in Proposition 3 and Proposition 4. The HJB argument in Proposition 3 implies that the government restricts exposure until cumulative exposure reaches $\widetilde{X}_a$ and grants full access thereafter. If the integral defining the threshold is no greater than one when cumulative exposure is zero, full access is optimal throughout. In that case, $-V'_a \leq 1$ everywhere, so full access is also optimal before an anticipated risky period.

When the government offers full exposure during hostility,

$$\frac{G_a'}{G_a} = -\mu - \frac{1}{C_Y^a - X},$$

while on the testing region,

$$G_a' < -\frac{G_a^2 r(r+\gamma)}{q_0 \gamma}.$$

Observe that these are the conditions used in the proof of Proposition 4, with the difference that C_Y replaced by C_Y^a. Thus, combined with generalized Assumption 1, they ensure that the single-crossing argument continues to hold and the characterization for the optimal exposure path remains the same albeit with generalized expressions.

□